VMSA: Vision Mamba With Spectral Attention for Hydrated Mineral Classification and Mapping From CRISM Hyperspectral Data

Yanchao Zheng, *Graduate Student Member, IEEE*, and Shuanggen Jin, *Senior Member, IEEE*

***Abstract*—Accurate identification and spatial mapping of hydrated minerals on Mars are fundamental for constraining ancient aqueous activity and reconstructing Martian environmental evolution. However, mineral recognition from orbital hyperspectral data remains challenging due to striping noise, weak diagnostic absorptions, complex background mixing, and limited labeled samples, which hinder classification accuracy and cross-scene generalization. To address these challenges, we propose a Vision Mamba with spectral attention (VMSA) framework for hydrated-mineral classification and mapping from Compact Reconnaissance Imaging Spectrometer for Mars (CRISM) hyperspectral data. VMSA adopts a coupled dual-branch architecture, in which a contextual spectral–spatial branch (CSB) learns stable spectral–spatial representations and a diagnostic-prior branch (DPB) explicitly models mineral-sensitive absorption priors, followed by joint discrimination through a prior-guided fusion (PGF) module. Based on public CRISM data resources, we further construct a unified hydrated-mineral benchmark and systematically evaluate the proposed method on three standardized subbenchmarks, namely, phyllosilicates, sulfates, and carbonates. Experimental results on the unified CRISM hydrated-mineral benchmark show that VMSA achieves strong and class-balanced performance across the phyllosilicate, sulfate, and carbonate subbenchmarks under a consistent evaluation protocol. In addition to benchmark-level quantitative comparison, we further conduct an endpoint-mined controlled input-space stress test to examine the behavior of the model under reduced hard-pair separability and diagnostic-prior perturbation. The results indicate that the diagnostic-prior pathway and PGF help to preserve target-versus-competing class margins when contextual/background cues become increasingly ambiguous. External evaluations on public hyperspectral image classification benchmarks, including Pavia University, Indian Pines, and Salinas, further support the transferability of the spectral–spatial backbone, while the CRISM-specific DPB remains dedicated to hydrated-mineral recognition. Finally, ratioed CRISM benchmark spectra were analyzed jointly with Mineral Identification and Classification Algorithm (MICA)-corrected reference spectra and laboratory spectra, while continuum-normalized band-depth contrasts were quantified for representative hard mineral pairs to assess the spectral discriminability and mineralogical consistency of VMSA predictions. These results suggest that VMSA provides a physically informed spectral–spatial framework for automated hydrated-mineral classification and mapping from CRISM data.**

***Index Terms*—Compact Reconnaissance Imaging Spectrometer for Mars (CRISM), deep learning, hyperspectral image, Martian mineral identification, spatial–spectral fusion features.**

Received 4 April 2026; revised 27 May 2026, 30 June 2026, and 5 July 2026; accepted 9 August 2026. Date of publication 11 August 2026; date of current version 25 August 2026. This work was supported by the National Key Research and Development Program of China under Grant 2021YFA0716100. *(Corresponding author: Shuanggen Jin.)*

Yanchao Zheng is with the School of Geomatics and Aerospace Information, Henan Polytechnic University, Jiaozuo 454003, China (e-mail: zycjy0430@163.com).

Shuanggen Jin is with the School of Geomatics and Aerospace Information, Henan Polytechnic University, Jiaozuo 454003, China, and also with the School of Artificial Intelligence, Anhui University, Hefei 230601, China (e-mail: sgjin@hpu.edu.cn; sg.jin@yahoo.com).

Digital Object Identifier 10.1109/TGRS.2026.3722719

I. Introduction

THE spatial distribution of hydrated minerals on Mars provides one of the most direct constraints on ancient aqueous activity and forms a key basis for reconstructing environmental evolution and assessing past habitability [1], [2], [3], [4]. The scientific value of orbital imaging spectroscopy lies not merely in indicating that liquid water once existed but in using mineral assemblages and co-occurrence relationships to constrain fluid chemistry, reaction pathways, and the duration and intensity of alteration [5], [6]. In recent years, Martian aqueous studies have gradually shifted from asking whether water was present to asking what kind of fluids existed and whether alteration proceeded through multiple stages. This shift makes species-level discrimination among spectrally similar hydrated minerals increasingly important for resolving water-rock interaction processes. This motivation is directly reinforced by the latest in situ results from Jezero Crater. Planetary Instrument for X-ray Lithochemistry (PIXL) measurements, the Mineral Identification by SToichiometry framework, and subsequent mineralogical analyses indicate that even within a limited stratigraphic setting, multiple alteration episodes and diverse mineral associations may coexist, and that robust paleoenvironmental interpretation often depends on mineral assemblages and minor components rather than on coarse orbital endmember labels alone [7], [8], [9].

Among currently available Martian orbital datasets, the Compact Reconnaissance Imaging Spectrometer for Mars (CRISM) remains the core hyperspectral data source for high-spatial-resolution hydrous mineral mapping, while the Observatoire pour la Minéralogie, l'Eau, les Glaces et l'Activité (OMEGA) and the Mars Mineralogical Spectrometer (MMS) provide useful cross-mission context and broader mineralogical comparisons [10]. However, robust mineral recognition from Martian hyperspectral data still faces well-documented, persistent challenges. First, heterogeneous preprocessing procedures, including atmospheric correction, destriping, bad-band handling, and reprojection, can modify

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

absorption depths, band center positions, and spectral curvature, thereby weakening reproducibility across scenes [11], [12], [13], [14]. Second, nonlinear spectral mixing within a basaltic background, dust cover, grain-size variation, and differences in hydration state can suppress or distort weak diagnostic absorptions, especially for low-abundance phases [15], [16]. Third, limited labeled samples, strong spatial autocorrelation, and complex background variability can encourage models to rely on spurious discriminative cues, thereby weakening cross-scene generalization performance [17], [18], [19]. In addition, because CRISM is inherently a pushbroom imaging spectrometer, image-extracted ratio spectra remain important under real scene conditions for validating weak mineral signals and mitigating the impact of residual column-dependent artifacts [14], [20].

Traditional Martian mineral mapping has mainly been built on physically meaningful spectral analysis frameworks. Representative approaches include CRISM summary products, diagnostic spectral parameters based on absorption depth and band shape, expert-guided browse product interpretation, analysis assisted by the CRISM Analysis Toolkit (CAT), spectral library matching via the Mineral Identification and Classification Algorithm (MICA), and manual inspection of raw and ratio spectra [12], [13], [14], [21], [22]. These approaches have played an essential role in the rapid screening and high-confidence identification of candidate hydrated-mineral outcrops and have underpinned numerous key discoveries in Martian alteration mineralogy. Their core strength lies in the direct link between spectral response and mineral-diagnostic absorptions, and they remain irreplaceable for expert validation and ground-truth confirmation workflows. However, such approaches usually depend on hand-crafted spectral parameters, empirical thresholds, and scene-dependent interpretation workflows. Their stability degrades substantially when target hydrated minerals exhibit weak diagnostic absorptions, occur in mixed pixels, have spatially scattered distributions, or are obscured by residual artifacts, rendering large-scale, consistent, and standardized mineral mapping highly labor- and time-intensive [6], [23].

To improve the degree of automation and consistency of mineral mapping, learning-based methods have been progressively introduced into CRISM mineral analysis tasks in recent years. Early machine learning frameworks integrated manually curated labels, spectral library information, and statistical classifiers to improve the scalability of automated mineral identification [21], [22], [23]. More recently, deep learning methods have advanced joint spectral–spatial representation learning for Martian hyperspectral data, demonstrating the strong potential of end-to-end models in handling complex backgrounds, subtle interclass spectral differences, and spatial contextual information [17], [24], [25], [26], [27], [28], [29], [30]. In particular, recent studies targeting Martian hyperspectral tasks have demonstrated that the key to improving mineral identification performance is shifting from single-pixel spectral discrimination toward more robust joint spectral–spatial modeling aligned with realistic application conditions. The multiscale spectral–spatial fusion network (MSSFNet) improves Martian mineral identification accuracy by combining multiscale spectral and spatial feature extraction with adaptive spectral–spatial fusion, while the mixed CNN–Transformer graph contrastive learning (MCTGCL) framework integrates a hybrid CNN–Transformer architecture with graph contrastive learning to enhance local feature representation and global contextual modeling capabilities [31], [32]. Meanwhile, sequential and task-oriented frameworks such as SHMD further reflect the growing demand to move beyond coarse endmember detection toward more structured hydrated mineral discrimination and quantitative abundance inversion analysis [33]. At the same time, explainable machine learning and open-set recognition have also been introduced to CRISM mineral mapping, marking a paradigm shift in the field from a sole focus on classification accuracy to deeper questions of model interpretability, classification reliability, and robustness when encountering unseen classes [27], [28]. Despite these notable advances, most existing methods still rely primarily on statistically driven discriminative feature learning and lack explicit structural-level modeling of the mineralogical priors embedded within the diagnostic absorption intervals of target hydrated minerals.

Despite the notable progress outlined above, several critical issues have not yet been fully addressed in current studies. First, most existing models still rely predominantly on statistically driven discriminative feature learning and have not explicitly integrated mineralogical spectral constraints derived from diagnostic absorption intervals into the representation learning process. As a result, their interpretability typically relies on post hoc validation, rather than being proactively embedded during the model design phase. Second, long-range spatial contextual modeling is commonly achieved either through attention mechanisms with high computational cost, or through stacked convolutional layers with inherently limited effective receptive fields. Third, across varying geological backgrounds and coexisting heterogeneous preprocessing pipelines, the cross-scene transferability of existing models remains unstable, and the mineralogical consistency of label annotations and prediction results still requires stronger support from image-extracted spectral evidence. These limitations collectively motivate the development of a methodological framework that integrates physically meaningful spectral priors with efficient long-range spatial contextual modeling.

To address these aforementioned challenges, this article proposes a Vision Mamba with spectral attention (VMSA) framework for species-level hydrated mineral identification and mapping from CRISM hyperspectral data. The proposed framework consists of a contextual spectral–spatial branch (CSB), a diagnostic-prior branch (DPB), and a prior-guided fusion (PGF) module. The CSB combines multiscale 3-D encoding, spectral attention modeling, and Mamba-based efficient context modeling to jointly enhance mineral-sensitive spectral–spatial representations and long-range spatial dependency learning. The DPB explicitly emphasizes mineral-diagnostic spectral information by constructing sample-adaptive interval responses and class-conditional interval modulation over mineral-sensitive wavelength ranges. The PGF module then integrates contextual features and prior-guided features into a unified representation. In this way,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

TABLE I
KEY PARAMETERS OF ORBITAL HYPERSPECTRAL SPECTROMETERS USED FOR MARTIAN MINERALOGICAL INVESTIGATIONS

Spectrometer	Orbiter	Launch time	Launch country	Spectral range/μm	Spectral resolution	Number of bands	Spatial resolution
TES	MGS	1996	NASA	5.8–50	10 cm^{-1} / 20 cm^{-1}	144 / 296	~3 × 6 km
OMEGA	MEX	2003	ESA	0.36–5.1	7–20 nm	352	300–4000 m
CRISM	MRO	2005	NASA	0.36–3.92	6.55 nm	544	18–200 m
MMS	TW-1	2020	CNSA	0.379–3.4	2.96–3.9 nm	576	265 m

the proposed framework is designed not only to improve quantitative classification performance but also to strengthen the mineralogical interpretability and geological credibility of automated mineral mapping results.

The main contributions of this work are summarized as three aspects.

1) The VMSA is proposed for Martian hydrated mineral identification, which integrates spectral attention, efficient Mamba-based long-range spatial contextual modeling, and a PGF mechanism. This design effectively improves the discrimination capability for spectrally similar mineral classes under conditions of weak diagnostic absorptions, mixed pixels, and limited labeled samples.
2) An explicit mineralogical prior modeling strategy centered on mineral diagnostic absorption intervals is developed. By integrating sample-adaptive interval responses with class-conditional interval modulation, the proposed method strengthens the correspondence between deep representation learning and mineral-sensitive spectral intervals, thereby achieving superior model interpretability compared with purely data-driven feature extraction approaches.
3) A systematic evaluation is conducted using a unified CRISM hydrated-mineral benchmark, endpoint-mined controlled input-space stress tests, and external public generic hyperspectral benchmarks. The CRISM benchmark remains the primary evidence for hydrated-mineral recognition, whereas the controlled stress test examines mechanism-level behavior under hard-pair perturbations and the public HSI benchmarks evaluate the transferability of the spectral–spatial backbone beyond CRISM. In addition, a spectral-consistency analysis was established by integrating ratioed CRISM spectra, MICA-corrected reference spectra, laboratory reference spectra, and pairwise continuum-normalized band-depth contrasts for representative hard mineral pairs.

II. RELATED WORKS

A. Mars Hyperspectral Datasets

Major orbital spectroscopic instruments used for Martian mineral investigations include the Thermal Emission Spectrometer (TES), OMEGA, CRISM, and MMS. Key specifications are summarized in Table I [2], [34], [35], [36]. TES operates mainly in the mid-infrared (5.5–50 μm) and is well suited for global-scale mapping of thermal inertia and anhydrous silicates, but its spatial resolution is relatively coarse at ~3 km/pixel. OMEGA covers the Visible and Near-InfraRed–ShortWave InfraRed (VNIR–SWIR) range at moderate spatial resolution and has been widely used for regional mapping of water ice and hydrated minerals because of its broad spatial coverage [1].

CRISM provides higher spatial resolution and more complete coverage of diagnostic absorption features for hydrated minerals. Its hyperspectral observations span 0.36–3.92 μm with 544 bands, and multiple observing modes are available to balance spatial resolution and areal coverage. Targeted observations reach tens of meters per pixel and enable fine-scale mineral mapping, whereas multispectral and mapping modes provide broader geological background constraints [14], [37]. Therefore, most studies on detailed mineral recognition focus on CRISM targeted data (e.g., full resolution targeted (FRT) and half resolution long (HRL) observations) and perform cross-scene comparison and generalization analysis after consistent preprocessing, including photometric correction, atmospheric correction, denoising, and reflectance normalization.

An important recent addition is MMS onboard China's Tianwen-1 mission. MMS is a pushbroom imaging spectrometer with an instantaneous field of view (IFOV) of ~0.5 mrad, covering 0.379–3.425 μm with 576 contiguous bands. It provides multiple in-orbit modes and band configurations to balance data volume and mineral detection requirements [10], [38]. Similar to CRISM, MMS may also suffer from systematic spectral shape biases. For example, in-orbit spectral smile can introduce spatial variations in center wavelength and full width at half maximum (FWHM), thereby reducing the consistency of reflectance retrieval and mineral identification [39].

B. Spectral Diagnostics of Martian Hydrated Minerals

Martian secondary mineralogy is highly diverse, and the chemical composition and crystal structure of different minerals control their absorption and scattering behavior in the VNIR–SWIR range, thereby producing diagnostic spectral features for remote mineral identification. For hydrated minerals on Mars, orbital recognition mainly relies on characteristic absorptions in the VNIR–SWIR region produced by combination vibrations of structural hydroxyl (OH), molecular water (H_2O), and metal–OH bonds [4], [39]. In general, absorptions near 1.4 μm and 1.9 μm mainly reflect hydroxyl and hydration information, whereas the position, bandwidth, relative depth, and shape organization of local absorption features in the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

2.1–2.5 μm range are usually more critical for fine-grained mineral discrimination.

Phyllosilicates commonly exhibit OH- and H_2O-related absorptions near 1.4 μm and 1.9 μm, whereas metal–OH absorptions in the 2.2–2.35 μm range are more diagnostic of mineral type [40], [41]. In general, Al–OH absorptions occur near 2.20 μm, whereas Fe–OH/Mg–OH absorptions are typically located in the 2.29–2.33 μm range, with subtle shifts in band center and shape caused by variations in cation composition and crystal structure. As a result, fine discrimination among phyllosilicate species often depends less on the presence of absorption features than on differences in local band position, bandwidth, and spectral shape organization within the 2.2–2.35 μm range.

In addition to the common hydration-related absorptions near 1.4 μm and 1.9 μm, sulfates exhibit a suite of diagnostic features in the 1.7–2.5 μm range that are jointly controlled by sulfate (SO_4) vibrational modes, structural water, and adsorbed water [42], [43], [44]. Sulfates with different hydration states often show systematic differences in absorption position and band shape. Therefore, discrimination between polyhydrated and monohydrated sulfates usually depends on the combined interpretation of multiple diagnostic intervals rather than a single absorption minimum. Although minerals such as jarosite and alunite both display distinct hydration- or hydroxyl-related absorptions, they exhibit identifiable differences in local spectral shape organization and relative depth patterns across adjacent 1.4–1.9 μm and 2.1–2.3 μm intervals, which provides robust evidence for orbital discrimination [44].

Carbonates typically show paired absorption features near 2.3 μm and 2.5 μm in the near-infrared range. Differences in local band position and relative depth are commonly used to distinguish $MgCO_3$ from $CaCO_3$/$FeCO_3$, and to further constrain paleofluid chemistry and local carbon cycle processes [45]. In orbital spectral interpretation for Mars, the structural organization of these paired absorptions is often more informative than individual band center values, especially under conditions of strong background mixing, attenuated absorptions, or severe spectral distortion.

Besides phyllosilicates, sulfates, and carbonates, hydrated silica and zeolite-group minerals are also of high importance in Martian orbital mineral interpretation. Hydrated silica commonly exhibits diagnostic Si–OH-related absorptions in the 2.21–2.23 μm range, often accompanied by a hydration-related absorption near 1.9 μm, with its band center and width varying with crystallinity and hydration state [46], [47]. Zeolite minerals such as analcime show hydration-related absorptions near 1.42 μm and 1.91 μm, together with localized diagnostic features near 1.79 μm and 2.52 μm. Consequently, their orbital identification usually requires joint examination of multiple bands rather than reliance on a single spectral feature [3], [48].

Mineral identification in Martian remote sensing commonly relies on the combined use of laboratory and planetary spectral libraries. The USGS Digital Spectral Library and the Reflectance Experiment Laboratory (RELAB) Spectral Library provide extensive VNIR–SWIR reflectance spectra of minerals and mineral mixtures, forming a critical basis for endmember selection and absorption feature attribution. The MICA library further compiles representative type spectra curated from peer-reviewed CRISM observations and published mineral identifications, providing a key reference for Martian orbital mineral interpretation [22], [49]. The taxonomy of Martian hydrated minerals used in the MICA spectral library is summarized in Table II. Traditional library-based interpretation usually performs mineral discrimination using combined spectral parameters including absorption depth, band center position, bandwidth, and spectral curvature. For CRISM data, the summary products framework further organizes this information into spectral parameter combinations suitable for rapid regional screening and comparative analysis [13], [14]. Nevertheless, factors including grain size variations, dust cover, areal and nonlinear spectral mixing, as well as atmospheric correction residuals, striping noise, and bad-pixel artifacts, can significantly attenuate or distort subtle diagnostic absorption features. As a result, pixel-level spectral similarity matching alone often leads to missed detections or misclassifications, particularly for low-abundance mineral phases and spectrally similar categories.

It should be clarified that the spectra shown in Fig. 1 are ratioed I/F reference spectra from the CRISM Type Spectra Library/MICA-related reference context. They are used only to illustrate representative diagnostic absorption behavior and to support mineralogical interpretation. They are not directly used as training samples for VMSA. The supervised benchmark used in this study is constructed from the public CRISM machine-learning resource, where labeled pixels are traced back to CRISM targeted observations through image identifiers, pixel coordinates, and connected-patch metadata. Therefore, the reference spectra in Fig. 1, the benchmark training samples, and the image-extracted spectra used for post hoc verification play different roles in this work.

C. Traditional and Learning-Based Methods for Martian Mineral Mapping

Early Martian orbital mineral mapping was mainly built on physically meaningful spectral analysis frameworks. Representative approaches include CRISM summary products, diagnostic spectral parameters derived from absorption depth and band shape characteristics, analysis assisted by CAT, expert-led browse product interpretation, spectral matching with MICA-type spectra, and manual inspection of raw and ratio spectra [13]. As these methods are directly rooted in the physics of mineral diagnostic absorption, they have played a pivotal role in the rapid screening and high-confidence identification of candidate mineral outcrops and have underpinned a large body of foundational research on Martian alteration mineralogy [6], [13], [14]. In parallel, efforts such as the Mars Orbital Catalog of Aqueous Alteration Signatures (MOCAAS) have systematically organized aqueous alteration signatures revealed by OMEGA and CRISM observations into a standardized orbital mineral catalog, providing critical mineralogical context from regional to near-global scales [6]. However, these approaches typically rely on hand-crafted spectral parameters, empirical thresholds, and site-dependent interpretation workflows. Their stability degrades substantially when target minerals exhibit weak absorptions, occur in mixed pixels, have spatially scattered distributions, or are obscured

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

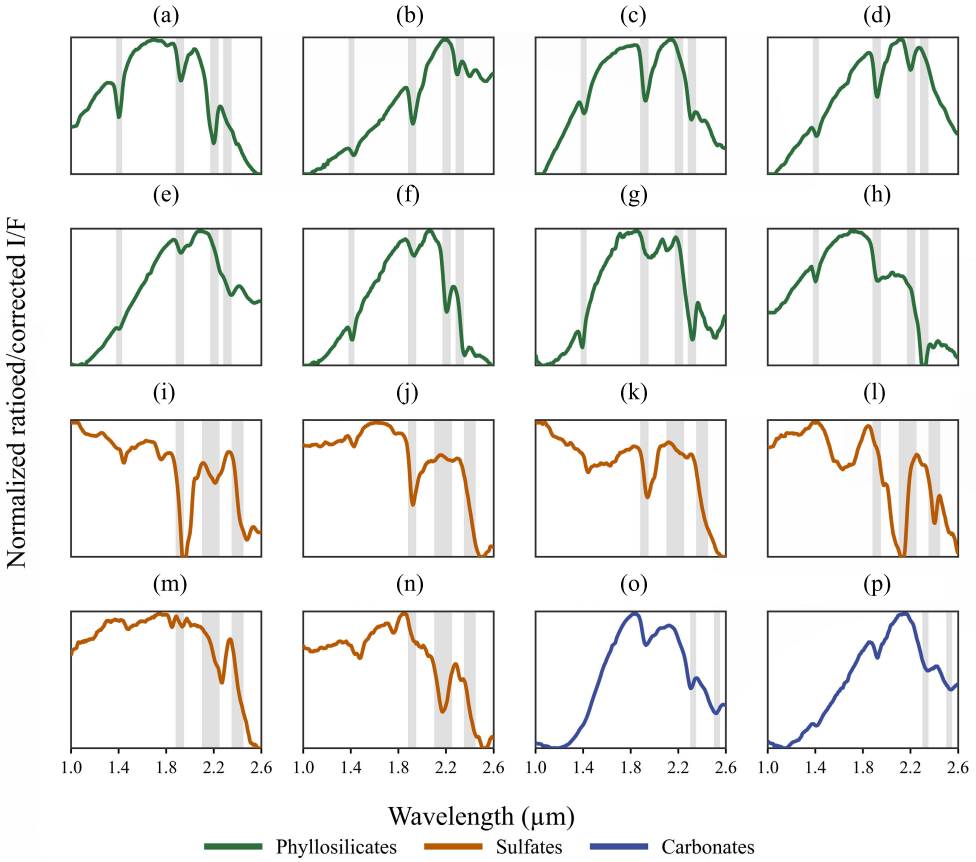


Fig. 1. Representative normalized ratioed/corrected I/F spectra of hydrated minerals from the CRISM type spectra library/MICA-related reference context. The spectra are shown only as diagnostic reference spectra for mineralogical interpretation and are not directly used as model training inputs. Gray shaded regions indicate representative wavelength intervals containing major hydration-, hydroxyl-, metal–OH-, sulfate-, and carbonate-related diagnostic absorptions. (a) Kaolinite. (b) Fe-smectite. (c) Mg-smectite. (d) Al-smectite. (e) Chlorite. (f) Illite/muscovite. (g) Serpentine. (h) Talc. (i) Gypsum. (j) Bassanite. (k) Polyhydrated sulfate. (l) Monohydrated sulfate. (m) Jarosite. (n) Alunite. (o) $MgCO_3$. (p) $CaCO_3/FeCO_3$.

by residual artifacts, rendering large-scale, consistent, and standardized mineral mapping highly labor-intensive.

To improve the degree of automation and cross-scene consistency, learning-based methods have been progressively introduced into CRISM mineral analysis. Early studies mainly combined manually curated labels, spectral library information, and statistical classifiers for automated mineral recognition. Subsequent frameworks began to leverage joint spectral–spatial representation learning to address complex backgrounds, class imbalance, and spatial autocorrelation issues [17], [18], [19]. On this basis, the CRISM machine learning toolkit developed by Plebani et al. [17] provided an explicit public foundation for supervised learning experiments and standardized data organization. More recent studies have further introduced explainable machine learning, open-set recognition, and Mars-specific deep learning architectures, indicating that the field is gradually shifting from a sole focus on classification accuracy to deeper questions of model interpretability, decision reliability, and robustness in the presence of unseen classes [27], [28]. For example, recent explainable machine learning work used tree-based models together with SHapley Additive exPlanations (SHAP) analysis to interpret mineral discrimination processes in CRISM data [28], while open-set studies have explored more robust mineral recognition when previously unseen mineral classes are present. At the same time, Mars-oriented deep networks including the MCTGCL framework, MSSFNet, and the SHMD framework further emphasized the critical role of joint spectral–spatial modeling for capturing local spectral details, global spatial context, and task-oriented hydrated mineral discrimination. Despite these notable advances, most existing methods still rely predominantly on statistically learned discriminative features, with insufficient explicit structural modeling of mineral-sensitive diagnostic absorption intervals. This key gap constitutes the core motivation for the DPB and PGF strategy proposed in this study.

III. Methodology

A. Overall Architecture

This section presents VMSA, a deep learning framework for hydrated-mineral classification and mapping from CRISM hyperspectral data under label-limited and cross-scene conditions. VMSA is designed to address three coupled challenges in Martian mineral mapping: 1) learning discriminative and transferable spectral–spatial representations from sparse and unevenly distributed labels; 2) efficiently capturing both local spectral–spatial details and long-range

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

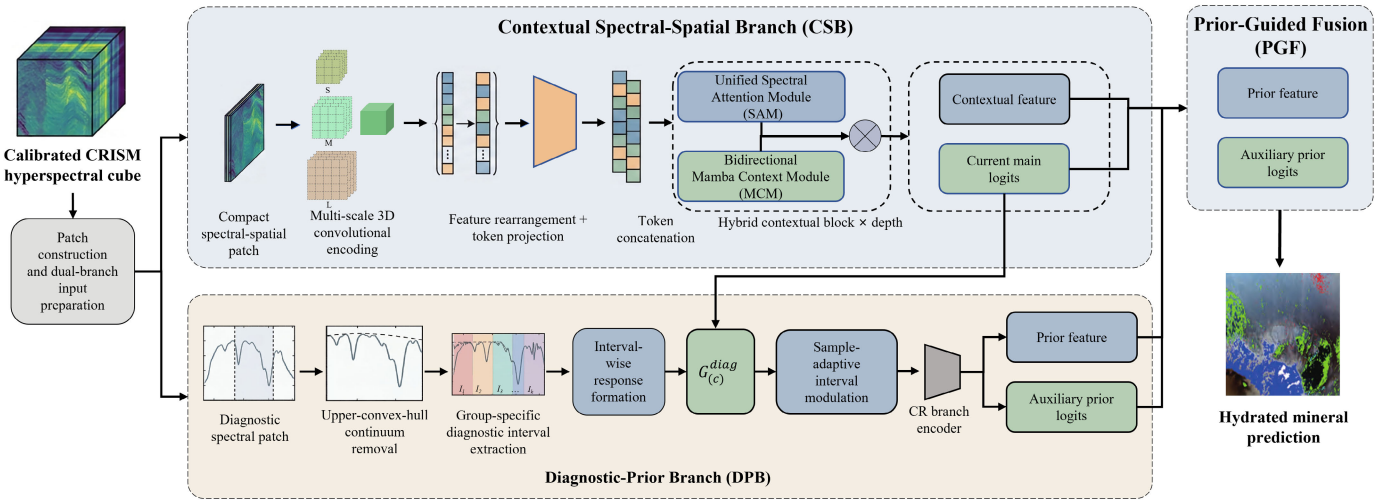


Fig. 2. Overall architecture of the VMSA framework.

contextual dependencies under high-dimensional observations and complex geological backgrounds; and 3) incorporating mineralogically motivated spectral priors into representation learning by explicitly modeling diagnostic wavelength ranges, continuum-removed absorption structures, and interval-level responses relevant to hydrated-mineral discrimination.

VMSA adopts a coupled dual-branch architecture. The first branch is a CSB, which operates on a compact spectral–spatial representation constructed from the consistently preprocessed CRISM scene cube. The CSB combines a multiscale 3-D encoder, a unified spectral attention module (SAM), and a bidirectional Mamba context module (MCM) to jointly model fine local spectral–spatial structures and broader contextual continuity. The second branch is a DPB, which operates on an upper convex-hull continuum-removed diagnostic-interval representation within mineral-diagnostic wavelength ranges. This branch decomposes the diagnostic spectra into predefined mineral-sensitive intervals and models interval-level responses through sample-adaptive interval gating and class-conditional interval modulation, thereby explicitly encoding absorption-structure information relevant to hydrated-mineral discrimination. The outputs of the two branches are subsequently integrated by a PGF module before classification.

Fig. 2 illustrates the overall architecture of VMSA. Let the consistently preprocessed CRISM hyperspectral cube be denoted by

$$X \in R^{D\times H\times W} \tag{1}$$

where D is the number of spectral channels and H and W are the spatial dimensions. For a labeled center pixel $p = (u, v)$, two complementary inputs are constructed through separate preprocessing protocols. The CSB input, denoted as X^p_{CSB}, is a compact spectral–spatial patch representation constructed from the preprocessed CRISM scene cube through the main-branch spectral input protocol, train-split standardization, and local patch extraction. The DPB input, denoted as X^p_{DPB}, is generated separately by diagnostic-range cropping, upper convex-hull continuum removal, and group-specific diagnostic-interval extraction.

Within this framework, the CSB and DPB play complementary roles. The CSB learns contextual spectral–spatial representations from X^p_{CSB} and focuses on multiscale local structures together with long-range spatial dependencies. In contrast, the DPB operates on X^p_{DPB} and explicitly emphasizes mineral-sensitive absorption structures through predefined diagnostic intervals, sample-adaptive interval gating, and class-conditional interval modulation. The final prediction for the center pixel is then obtained by

$$\hat{y}_p = f_{cls}\left(f_{PGF}\left(f_{CSB}\left(X^p_{CSB}\right), f_{DPB}\left(X^p_{DPB}\right)\right)\right) \tag{2}$$

where $f_{CSB}(\cdot)$ and $f_{DPB}(\cdot)$ denote the contextual and diagnostic-prior feature extractors, respectively, $f_{PGF}(\cdot)$ denotes the PGF function, and $f_{cls}(\cdot)$ denotes the final classification head.

B. Contextual Spectral–Spatial Branch

The CSB is designed to learn contextual spectral–spatial representations from CRISM observations. As shown in Fig. 3, the CSB takes compact spectral–spatial input patches as input, performs multiscale 3-D spectral–spatial encoding, and then models complementary contextual dependencies through the unified SAM and the bidirectional MCM. The resulting contextual features are fused and globally pooled to produce the contextual representation used for final classification.

For each labeled center pixel p, the CSB input is constructed from the consistently preprocessed CRISM scene cube through a fixed main-branch spectral input protocol, followed by train-split standardization and local patch extraction. This process can be written as

$$X^p_{\mathrm{CSB}} = Patch_s\left(N_{\mathrm{tr}}\left(\Phi_{\mathrm{CSB}}\left(X\right)\right), p\right) \tag{3}$$

where X denotes the consistently preprocessed CRISM scene cube, $\Phi_{\mathrm{CSB}}(\cdot)$ denotes the fixed main-branch spectral input construction protocol, $N_{\mathrm{tr}}(\cdot)$ denotes the standardization using statistics estimated only from the training split, and $Patch_s(\cdot)$ extracts a local patch centered at p. In this study, s is set to 11. This compact representation is used to provide broad spectral–spatial context for the CSB and is constructed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

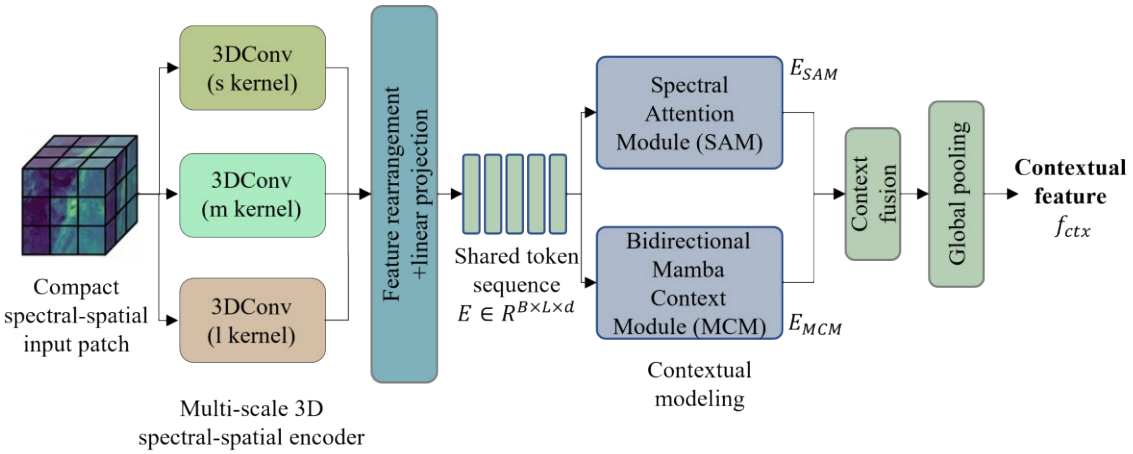


Fig. 3. Architecture of the CSB.

independently of the mineral-diagnostic interval representation used by the DPB.

For mini-batch training, the CSB input patches extracted from different center pixels are stacked along the batch dimension to form the input tensor X_{CSB}

$$X_{CSB} \in R^{(B\times 1\times C_m\times H_p\times W_p)} \tag{4}$$

where B denotes the batch size, C_m denotes the number of compact main-branch spectral channels, and H_p and W_p denote the spatial patch size. In this study, $H_p = W_p = 11$.

To characterize mineral spatial patterns at different scales, X_{CSB} is fed into a multiscale 3-D spectral–spatial encoder. Three parallel 3-D convolution branches with different receptive fields are used to extract spectral–spatial features from local to broader scales

$$F_k = \sigma\left(\mathrm{BN}\left(\mathrm{Conv}_{3\mathrm{D}}^{(K_k)}\left(X_{CSB}\right)\right)\right), \quad k = 1, 2, 3 \tag{5}$$

where K_k denotes the kernel size of the kth convolutional branch, BN$(\cdot)$ denotes the batch normalization, and $\sigma(\cdot)$ denotes the nonlinear activation function. With this multiscale parallel encoding strategy, the model can simultaneously capture fine-grained local details, neighborhood structural features, and broader spatial context, providing a multiscale feature foundation for subsequent contextual modeling.

The obtained multiscale feature tensors are then rearranged and projected into a shared token embedding space

$$E = \mathrm{Proj}\left(\mathrm{Rearrange}\left(F_1, F_2, F_3\right)\right) \tag{6}$$

where $E \in R^{B\times L\times d}$, L denotes the length of the token sequence, and d denotes the token embedding dimension. This operation unifies the 3-D spectral–spatial features into a token sequence suitable for subsequent sequential contextual modeling.

First, the unified SAM, whose internal structure is shown in Fig. 4, is used to model global interactions among spectral–spatial tokens

$$E_{SAM} = SAM\left(E\right) \tag{7}$$

where SAM consists of multihead self-attention, feed-forward refinement, and channel recalibration. This module extracts

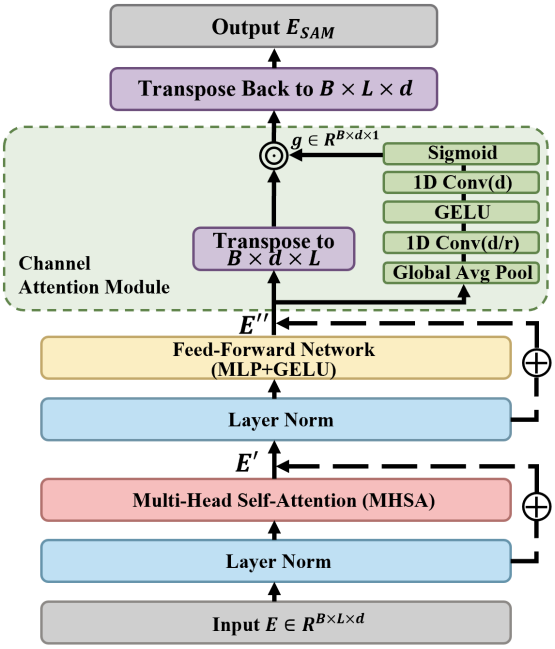


Fig. 4. Unified SAM.

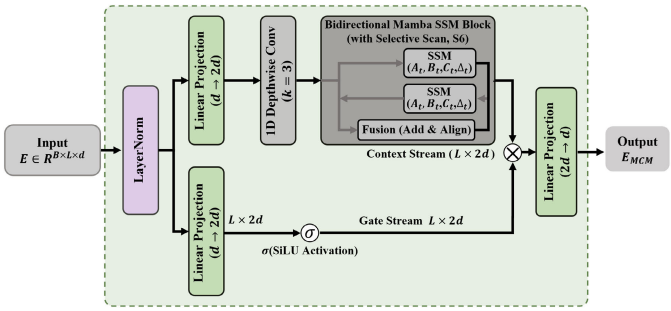


Fig. 5. Bidirectional MCM.

more discriminative contextual dependencies from global token interactions rather than relying only on local convolution responses.

Meanwhile, the bidirectional MCM, whose architecture is shown in Fig. 5, models long-range structured context

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

via selective state-space modeling. Its state update can be written as

$$h_t = A_t h_{t-1} + B_t e_t, \quad y_t = C_t h_t \tag{8}$$

where A_t, B_t, and C_t denote the discretized state transition terms. To reduce directional bias introduced by unidirectional scanning, a bidirectional scanning strategy is adopted, yielding

$$E_{\mathrm{MCM}} = \mathrm{BiMamba}(E). \tag{9}$$

Finally, the two contextual representations are fused to obtain the output feature of CSB

$$f_{ctx} = \mathrm{Pool}(E_{SAM} + E_{MCM}) \tag{10}$$

where Pool(·) denotes the global pooling along the token dimension. The resulting f_{ctx} jointly encodes local spectral–spatial structures and broader contextual continuity and is subsequently fused with the diagnostic-prior feature f_{prior} in the PGF module for final classification.

C. Diagnostic-Prior Branch

The DPB is designed to explicitly model diagnostic absorption structures relevant to hydrated-mineral recognition, rather than relying solely on the spectral–spatial discriminative features learned by the contextual branch. Unlike the CSB, which primarily focuses on learning stable spectral–spatial context and long-range spatial continuity, the DPB directly operates on spectral inputs within the mineral-diagnostic wavelength range. Through upper convex-hull continuum removal, group-specific diagnostic interval extraction, sample-adaptive interval response modeling, and class-conditional interval modulation, the DPB introduces mineral-sensitive absorption-structure priors into the feature-learning process.

For a labeled center pixel p, the diagnostic-prior input is constructed from the corresponding spectral patch through a preprocessing pathway independent of the CSB input. The spectral patch is first cropped to the wavelength range where hydrated-mineral diagnostic absorptions are concentrated. An upper convex-hull continuum is then constructed for each diagnostic spectrum, and the continuum-removed reflectance is computed as

$$R_c(\lambda) = \frac{R(\lambda)}{C_{\mathrm{hull}}(\lambda)} \tag{11}$$

where $R(\lambda)$ denotes the cropped reflectance spectrum, $C_{\mathrm{hull}}(\lambda)$ denotes the upper convex-hull continuum interpolated over the diagnostic wavelength grid, and $R_c(\lambda)$ denotes the continuum-removed reflectance. This operation suppresses broad albedo and spectral-slope variations while preserving local absorption morphology. Unlike a simple endpoint-line normalization, the upper convex-hull continuum follows the upper envelope of the diagnostic spectrum and is therefore more appropriate for emphasizing absorption structures in CRISM spectral verification and diagnostic-prior modeling.

The diagnostic-prior input for DPB is then written as

$$X_{DPB}^{p} = \mathrm{Extract}_{I_g}\left(R_c^{p}\right) \tag{12}$$

where X_{DPB}^{p} denotes the diagnostic-prior patch used by the DPB, R_c^{p} denotes the upper convex-hull continuum-removed spectral patch centered at p, and $\mathrm{Extract}_{I_g}(\cdot)$ denotes the diagnostic-interval extraction using the interval set I_g defined for the corresponding mineral group g. In this study, g refers to one of the three hydrated-mineral subbenchmarks: phyllosilicates, sulfates, and carbonates.

For each mineral group g, the diagnostic interval set is defined as

$$I_g = \left\{ I_k = \left[\lambda_k^s, \lambda_k^e\right] \;\middle|\; k = 1, 2, \ldots, K_g \right\} \tag{13}$$

where λ_k^s and λ_k^e denote the start and end wavelengths of the kth diagnostic interval, respectively, and K_g is the number of diagnostic intervals used for mineral group g. These intervals are fixed at the group level and are defined according to hydration-, hydroxyl-, metal–OH-, sulfate-, and carbonate-sensitive wavelength regions commonly used in CRISM mineral interpretation. They are not learned spectral templates for individual samples, nor are they the compact channels used by the CSB. Instead, they provide a physically motivated interval structure within which the DPB learns sample-adaptive and class-conditional diagnostic responses. For each diagnostic interval I_k, an interval-specific descriptor is extracted as

$$z_k = E_k\left(X_{DPB}^{p} \middle| I_k\right), \quad k = 1, 2, \ldots, K_g \tag{14}$$

where $X_{DPB}^{p}|I_k$ denotes the subrepresentation of the diagnostic-prior input within interval I_k, $E_k(\cdot)$ denotes the encoder for the kth interval, and z_k is the corresponding interval-level feature descriptor. Unlike encoding the entire diagnostic spectrum as a single holistic vector, this intervalwise design allows local absorption responses from different diagnostic wavelength ranges to be modeled separately. Based on these interval descriptors, sample-adaptive interval response weights are learned as

$$\alpha_k = G_k(z_k), \quad \tilde{z}_k = \alpha_k \odot z_k \tag{15}$$

where $G_k(\cdot)$ denotes a lightweight gating function, α_k denotes the adaptive response weight of the current sample on the kth diagnostic interval, $\odot$ denotes the elementwise multiplication, and $\tilde{z}_k$ denotes the weighted interval feature. This step enables the model to adaptively adjust the contributions of different diagnostic intervals according to the specific spectral shape of the current sample, which is particularly important under weak absorptions, locally overlapping band shapes, and complex background-induced spectral variations.

To further connect interval responses with class discrimination, class-conditional interval preference is introduced on top of the sample-adaptive interval responses. Let C_g be the number of mineral classes in group g. The interval preference of class c is defined as

$$P_c = \left[p_{c,1}, p_{c,2}, \ldots, p_{c,K_g}\right], \quad c = 1, 2, \ldots, C_g \tag{16}$$

where $p_{c,k}$ denotes the relative preference of class c for the kth diagnostic interval. Based on this class-conditional preference, the prior response for class c is written as

$$r_c = \sum_{k=1}^{K_g} p_{c,k} \tilde{z}_k. \tag{17}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

This operation does not impose rigid fixed mineral templates. Instead, it allows the model to learn relative interval sensitivities for different mineral classes under a unified training framework, thereby linking class discrimination with diagnostic absorption structures in an explicit but flexible manner.

Finally, the prior-guided feature produced by the DPB is written as

$$f_{\text{prior}} = F_{\text{prior}}\left(\left[r_1; r_2; \ldots; r_{C_g}\right]\right) \tag{18}$$

where [;] denotes the feature concatenation and $F_{\text{prior}}(\cdot)$ denotes the prior-feature projection module. The resulting f_{prior} is subsequently fused with the contextual feature f_{ctx} generated by the CSB in the PGF module for final classification. In addition to the prior-guided feature used for classification, the DPB provides interpretable interval-level quantities, including sample-adaptive interval responses and class-conditional interval preferences. These quantities are used in Sections IV and V to analyze diagnostic-interval behavior and to connect model decisions with image-extracted raw I/F, same-column ratioed I/F spectra, and upper convex-hull continuum-removed spectra.

D. PGF and Optimization

In the proposed VMSA framework, the contextual feature f_{ctx} produced by the CSB and the prior-guided feature f_{prior} produced by the DPB play complementary roles. The former captures local spectral–spatial structures and long-range contextual continuity, whereas the latter emphasizes diagnostic absorption structures relevant to hydrated-mineral discrimination. To jointly exploit these two sources of information, a PGF module is introduced to integrate them and generate the final discriminative representation for classification. The fusion process is formulated as

$$f_{fuse} = F_{PGF}\left(f_{ctx}, f_{prior}\right) \tag{19}$$

where $F_{\text{PGF}}(\cdot)$ denotes the PGF module, and f_{fuse} denotes the fused feature representation. The classification head then produces the class-probability vector for the center pixel p

$$p_p = \text{Softmax}\left(\text{Cls}\left(f_{\text{fuse}}\right)\right) \tag{20}$$

where $Cls(\cdot)$ denotes the classification mapping function, p_p denotes the class-probability vector for center pixel p, and $p_{p,c}$ denotes the predicted probability that pixel p belongs to class c. The predicted class label $\hat{y}_p$ is obtained as

$$\hat{y}_p = argmax_c p_{p,c}. \tag{21}$$

At the optimization stage, VMSA is trained with a joint objective composed of the main classification loss, an interval-consistency constraint, and a pairwise-margin constraint

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_1 \mathcal{L}_{\text{ic}} + \lambda_2 \mathcal{L}_{\text{pm}} \tag{22}$$

where $\mathcal{L}_{\text{cls}}$ is the main classification loss, $\mathcal{L}_{\text{ic}}$ encourages consistency between sample-level interval responses and class-conditional interval preferences, and $\mathcal{L}_{\text{pm}}$ enhances a positive logit margin between predefined hard mineral pairs. The coefficients λ_1 and λ_2 weight the interval-consistency and pairwise-margin constraints relative to the main classification loss, respectively, while m denotes the target logit margin used in L_{pm}. In all final experiments, λ_1, λ_2, and m are fixed to 0.04, 0.025, and 0.2, respectively, and are not tuned separately for different mineral groups.

The main classification loss is formulated as a class-reweighted cross-entropy loss to alleviate the adverse impact of class imbalance in the hydrated-mineral benchmarks [52]

$$\mathcal{L}_{\text{cls}} = -\frac{1}{B}\sum_{b=1}^{B} w_{y_b} \log p_{b,y_b} \tag{23}$$

where B denotes the batch size, y_b denotes the ground-truth label of the bth sample, p_{b,y_b} denotes the model's predicted probability for the ground-truth class, and w_{y_b} denotes the class-rebalancing weight for class y_b, calculated based on the inverse class frequency of the training set. This term maintains the core classification objective under the unified training framework, while mitigating the bias toward majority classes caused by imbalanced sample distribution.

To encourage the interval responses learned by the DPB to remain consistent with class-level diagnostic preferences, an interval-consistency constraint is further introduced. Let the sample-level interval response vector of the bth sample be

$$a^{(b)} = \left[\alpha_1^{(b)}, \alpha_2^{(b)}, \ldots, \alpha_{K_g}^{(b)}\right] \tag{24}$$

where $\alpha_K^{(b)}$ is the sample-adaptive interval weight of the bth sample on the kth diagnostic interval. Let the ground-truth class of the bth sample be y_b, and let the corresponding class-conditional interval preference vector be

$$P_{y_b} = \left[p_{y_b,1}, p_{y_b,2}, \ldots, p_{y_b,K_g}\right]. \tag{25}$$

The interval-consistency loss is then formulated as

$$\mathcal{L}_{ic} = \frac{1}{B}\sum_{b=1}^{B} \left|\left|a^{(b)} - P_{y_b}\right|\right|_2^2. \tag{26}$$

This term does not impose rigid fixed mineral templates. Instead, under the unified multiclass training framework, it encourages the model to learn sample-level interval response patterns that are consistent with class-conditional diagnostic preferences, further strengthening the explicit link between the model's discriminative features and mineral-sensitive absorption structures.

Considering that multiple hydrated mineral classes exhibit highly similar diagnostic absorption positions and spectral shapes, a pairwise-margin constraint is further introduced to improve fine-grained discrimination between hard mineral pairs. For samples belonging to predefined spectrally confusing mineral pairs, let o_{b,y_b} be the classification logit of the ground-truth class, and o_{b,c_b^-} be the logit of the most confusing competing class that forms a predefined hard pair with y_b. The pairwise-margin loss is defined as [53]

$$\mathcal{L}_{\text{pm}} = \frac{1}{|H|}\sum_{b\in H} max\left(0, m - \left(o_{b,y_b} - o_{b,c_b^-}\right)\right) \tag{27}$$

where H denotes the set of samples in the mini-batch that belong to predefined hard mineral pairs, $|H|$ is the number of such samples, m denotes the positive margin, and c_b^- denotes the competing class paired with y_b. This constraint explicitly enlarges the decision margin between spectrally

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

similar mineral classes, which directly addresses the fine-grained hydrated-mineral discrimination problem considered in this study.

In the final implementation, the CSB uses compact spectral–spatial input patches with a spatial size of 11×11, while the DPB uses group-specific upper convex-hull continuum-removed diagnostic-interval representations. The token embedding dimension is set to $d = 32$, and the depth of the contextual backbone is set to $T = 3$. The CSB, DPB, PGF, and classification head are jointly optimized in an end-to-end manner.

E. Evaluation Metrics

To assess the effectiveness of the proposed method, we utilize several key performance indicators: overall accuracy (OA), average accuracy (AA), precision (Pr), recall (Rc), $F1$-score ($F1$), and Kappa Coefficient (κ). These metrics provide a comprehensive evaluation of the model's performance from different perspectives.

OA: This is the most intuitive metric, representing the ratio of correctly classified pixels to the total number of test pixels. It provides a general measure of the model's performance across all classes. The formula is

$$OA = \frac{\sum_{i=1}^{C} TP_i}{\text{Total Number of Pixels}} \tag{28}$$

where true positives (TP_i) is the number of pixels correctly assigned to class i, and C is the total number of classes.

AA: To counteract the potential bias of OA in datasets with imbalanced class sizes, we calculate the AA. It is the mean of the classification accuracies for each individual class, treating each class with equal importance. It is defined as

$$AA = \frac{1}{C} \sum_{i=1}^{C} \frac{TP_i}{TP_i + FN_i} \tag{29}$$

where false negatives (FN_i) is the number of pixels belonging to class i but misclassified as another class.

Precision (Pr), Recall (Rc), and $F1$-Score: For a more granular, per-class analysis, we compute these three metrics.

Precision measures the proportion of TPs among all instances predicted as that class, indicating the reliability of the positive predictions

$$Precision = \frac{TP}{TP + FP} \tag{30}$$

where false positives (FPs) are pixels incorrectly assigned to the class.

Recall measures the proportion of TPs that were correctly identified, indicating the model's ability to find all relevant instances of a class

$$Recall = \frac{TP}{TP + FN}. \tag{31}$$

The $F1$-score is the harmonic mean of precision and recall, providing a single score that balances the tradeoff between making FP and FN errors

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{32}$$

TABLE II
TAXONOMY OF MARTIAN HYDRATED MINERALS USED IN THE MICA SPECTRAL LIBRARY

Category	Mineral	Key diagnostic absorption features (μm)
Phyllosilicates	Kaolinite	1.39-1.41, 2.16-2.21
	Fe/Mg-smectite	1.39-1.41 , 1.90-1.92 , 2.29-2.32
	Al-smectite	1.39-1.41 , 1.90-1.92 , 2.20-2.21
	Chlorite	1.39-1.41 , 2.25, 2.33-2.36
	Illite/Muscovite	1.4, 1.9, 2.20-2.21, 2.34-2.36, 2.44-2.46
	Serpentine	1.39, 2.10-2.12, 2.31-2.33, 2.50-2.52
	Talc	1.39, 2.31, 2.39, 2.46
Sulfates	Gypsum	1.44-1.53, 1.74-1.75, 1.94, 2.17, 2.21, 2.27
	Bassanite	1.4, 1.9, 2.1-2.4
	Polyhydrated sulfates	1.4, 1.9, 2.4
	Monohydrated sulfates	2.1, 2.4
	Jarosite	1.47, 1.85, 2.26-2.27
	Alunite	1.43, 1.48, 1.76, 2.16-2.17
Carbonates	Mg CO_3	2.30-2.31, 2.50-2.52
	Ca/Fe CO_3	2.33-2.35, 2.53-2.54

Kappa Coefficient (κ): This statistic provides a robust measure of how much better the classification is compared with a random assignment, by correcting for chance agreement. It is widely used in remote sensing for assessing classification accuracy. A value of 1 indicates perfect agreement, while a value of 0 indicates that the agreement is no better than random chance.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Datasets and Implementation Details

In this study, we construct a unified benchmark for hydrated-mineral recognition based on the public CRISM machine-learning resource released by Plebani et al. [17], rather than using temporarily assembled samples or manually selected pixels from individual scenes. This public CRISM resource comprises a ratioed labeled-pixel library, bland-pixel records, image identifiers, pixel coordinates, and connected-patch metadata. These metadata provide a reproducible basis for benchmark construction, spatially disjoint training/validation/testing, pixel-level mapping, and scene-level spectral analysis.

For the task of hydrated-mineral recognition on Mars, we build a standardized benchmark for hydrated-mineral

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

TABLE III
CLASS COMPOSITION AND TRAIN/VALIDATION/TEST STATISTICS OF THE UNIFIED HYDRATED-MINERAL BENCHMARK USED IN THIS STUDY

Mineral group	Mineral class	Train	Val	Test	Total
Carbonates	Ca/Fe CO_3	5320	1176	764	7260
	Mg CO_3	23505	2527	10557	36589
	Subtotal	28825	3703	11321	43849
Sulfates	Jarosite	8224	1086	3018	12328
	Alunite	4731	1649	1698	8078
	Polyhydrated sulfate	21130	10145	15369	46644
	Monohydrated sulfate	19887	1747	4317	25951
	Subtotal	53972	14627	24402	93001
Phyllosilicates	Fe smectite	55225	7569	12491	75285
	Mg smectite	86554	5214	17674	109442
	Beidellite	5954	514	2154	8622
	Kaolinite	22955	3325	8366	34646
	Illite	2235	680	1552	4467
	Chlorite	40147	2303	10567	53017
	Subtotal	213070	19605	52804	285479
	Total	**295867**	**37935**	**88527**	**422329**

classification from CRISM data, covering three major hydrated-mineral groups: phyllosilicates, sulfates, and carbonates. On this basis, public candidate samples are further screened, reorganized, and consistently preprocessed. Only representative classes that support stable supervised classification are retained, forming the standardized dataset used for subsequent training, validation, and testing. The final class composition and corresponding train/validation/test statistics are listed in Table III.

It should be emphasized that the labels in the public resource are treated as candidate mineral annotations rather than unquestioned absolute ground truth. To obtain an evaluation-ready supervised benchmark, nonhydrated mineral entries, obvious artifact-related samples, and classes with insufficient support for stable supervised learning are first excluded. The remaining samples are then re-examined by jointly considering the published mineral labels, source image identifiers, connected-patch indices, pixel coordinates, and spatial coherence of candidate regions. The retained mineral classes were further examined by comparing class-level ratioed CRISM benchmark spectra with MICA-corrected and laboratory reference spectra. Pairwise continuum-normalized band-depth contrasts were additionally quantified for representative hard mineral pairs to evaluate the consistency between spectral discrimination and expected mineralogical absorption behavior. This screening procedure is intended to improve the reliability of the benchmark while preserving traceability to the original public CRISM resource.

To illustrate representative geological backgrounds covered by the dataset, six CRISM targeted observations were selected from the full set of source scenes as scene-level examples in the main text, including FRT00009326 and FRT0000C467 in Mawrth Vallis, FRT0000A09C and FRT00009971 in Nili Fossae, and FRT00013EEF and FRT0000750A in Columbus Crater. Their geographic locations and regional geological contexts are shown in Fig. 6. Fig. 7 presents the spectral context of the 12 hydrated-mineral classes through a three-tier comparison of ratioed CRISM benchmark spectra, MICA-corrected

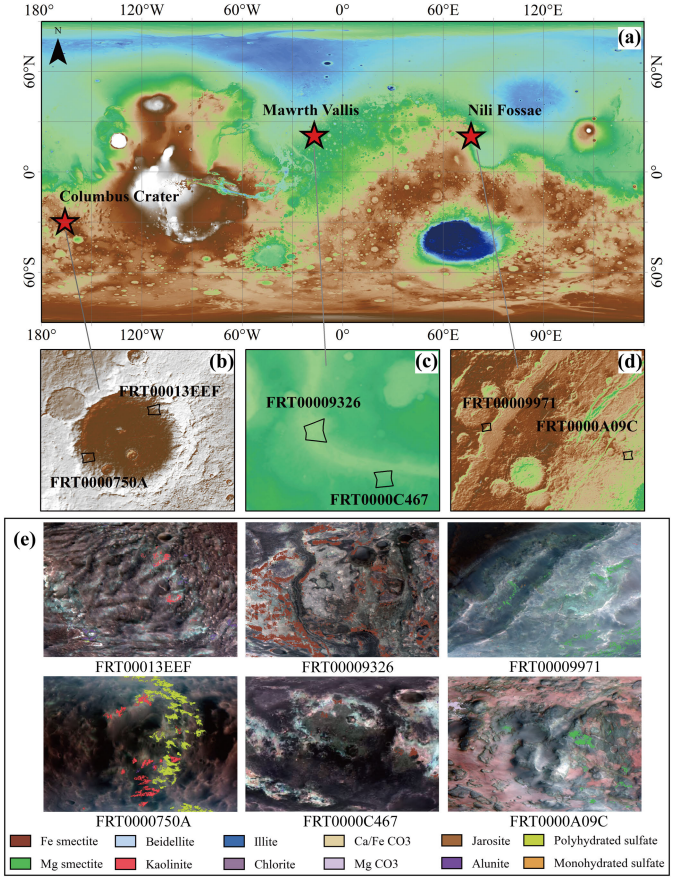


Fig. 6. Geographic locations and representative scene-level examples of the six CRISM targeted observations used for scene-level illustration. (a) Global distribution of the three representative hydrated-mineral regions on Mars. (b)–(d) Regional context maps for Columbus Crater, Mawrth Vallis, and Nili Fossae, respectively, shown on the MOLA DEM background. (e) False-color scene examples of the six representative CRISM observations, generated from the processed CRISM TRR3 cubes used in this study, with unified-benchmark mineral labels overlaid for qualitative reference. All regional-context panels are displayed on the MOLA DEM background.

type-reference spectra, and laboratory reflectance analogs. The panel titles retain the broad observation-based CRISM

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

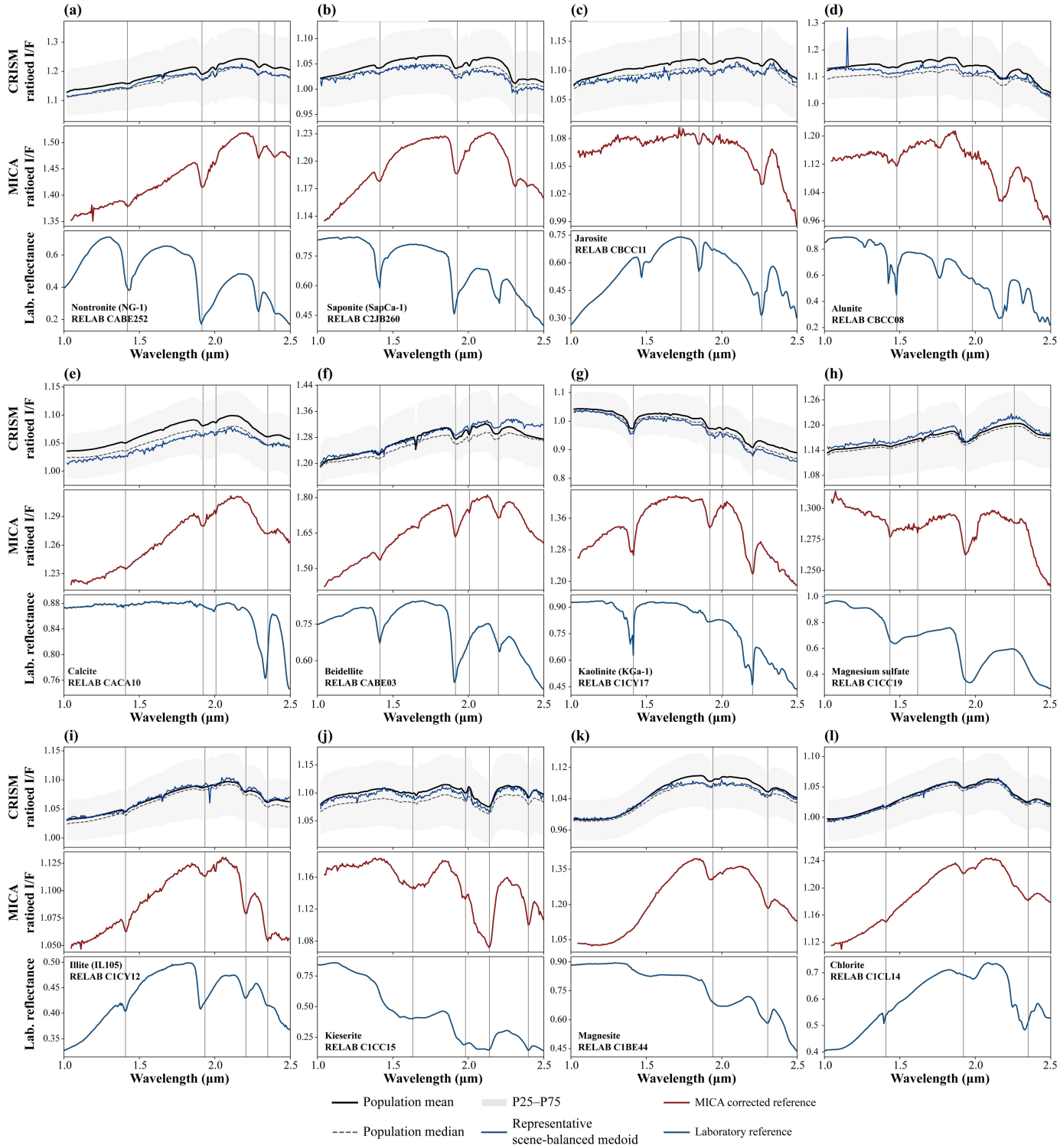


Fig. 7. Class-level spectral context for the 12 hydrated-mineral classes used in the unified benchmark. Within each panel, the top row shows ratioed CRISM benchmark spectral statistics, including the population mean, interquartile range, population median, and a representative scene-balanced medoid spectrum; the middle row shows the corresponding MICA-corrected CRISM type-reference spectrum; and the bottom row shows the selected laboratory reflectance analog. (a) Fe-smectite. (b) Mg-smectite. (c) Jarosite. (d) Alunite. (e) $CaCO_3/FeCO_3$. (f) Beidellite. (g) Kaolinite. (h) Polyhydrated sulfate. (i) Illite. (j) Monohydrated sulfate. (k) $MgCO_3$. (l) Chlorite.

benchmark classes, whereas the middle-row curves provide Mars-specific MICA type-spectrum context and the bottom-row curves represent specific terrestrial laboratory specimens. The mineral types of the laboratory analogs were cross-checked against the Similar Laboratory Spectra associations of the corresponding MICA type spectra, and each plotted laboratory curve is annotated with its specific analog mineral name and original RELAB spectrum identifier [22].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

These laboratory spectra provide traceable diagnostic reference context for the broad CRISM classes rather than one-to-one mineralogical assignments for all pixels within each class.

All source scenes are based on CRISM Targeted Reduced Data Record Version 3 (TRR3) products, and a unified preprocessing workflow is applied using the CAT and its associated ancillary files, including photometric correction, atmospheric correction, and related geometric processing, to ensure inputs are as consistent as possible across scenes in the reflectance domain [37], [54]. Regions affected by shadows, edge contamination, bad detectors, and low-SNR artifacts are masked out before sample extraction. To avoid inconsistencies between displayed imagery and actual model inputs, the false-color background images shown in the main text are all generated from the CAT-processed TRR3 cubes used in actual training and analysis, rather than directly taken from browse-style Merged Targeted Reduced Data Record (MTRDR) products.

Preprocessing is performed separately according to the dual-branch design of VMSA. For the CSB, compact spectral–spatial input patches are constructed from the consistently preprocessed CRISM scene cube through a fixed main-branch spectral input protocol, train-split standardization, and local patch extraction. The spatial patch size is set to 11×11. The CSB input is used to learn contextual spectral–spatial representations and is constructed independently of the mineral-diagnostic interval representation used by the DPB.

For the DPB, the corresponding spectra are processed through diagnostic-range cropping, upper convex-hull continuum removal, and group-specific diagnostic-interval extraction. The upper convex-hull continuum is constructed as the upper envelope over the diagnostic wavelength grid, and the continuum-removed reflectance is used to emphasize local absorption morphology while reducing broad albedo and spectral-slope effects. In this way, the CSB provides compact contextual spectral–spatial representations, whereas the DPB explicitly models mineral-sensitive absorption structures through diagnostic-interval priors.

To avoid spatial leakage and overly optimistic evaluation caused by random pixel-level splitting, a spatially disjoint partition protocol is adopted to construct the training, validation, and test sets. During training, class-weighted cross entropy is used to alleviate class imbalance, and model parameters are optimized using AdamW. Unless otherwise stated, the initial learning rate is set to 1×10^{-3}, the batch size is set to 64, and the maximum number of training epochs is set to 150. The final model is selected according to validation performance and evaluated on the held-out test set. All experiments are implemented in PyTorch 2.2.2 with CUDA 11.8 and conducted on a workstation equipped with an Intel Core i7-12600KF CPU and an NVIDIA GeForce RTX 4060 GPU with 8-GB memory.

For fair comparison and reproducibility, all methods are evaluated under identical training/validation/test splits, the same benchmark-level preprocessing derived from the CAT-processed CRISM cubes, and the same evaluation metrics. Method-specific input formatting is applied only to satisfy the architectural requirements of each model, without changing the data split, labels, preprocessing protocol, or test-set evaluation. No broad model-specific hyperparameter search is performed. Instead, all experiments follow a fixed protocol with a random seed of 0, and the checkpoint with the highest validation Macro-$F1$ is used for final test-set evaluation.

B. Quantitative Comparison on Standardized CRISM Benchmarks

To quantitatively evaluate the performance of the proposed VMSA framework under a unified evaluation protocol, we systematically compare VMSA with several representative state-of-the-art spectral–spatial hyperspectral classification baselines, including A^2S^2K-ResNet [55], SpectralFormer [30], MCTGCL [32], MSSFNet [31], SSFTT [56], and MambaHSI [57]. All methods are evaluated on the same three standardized CRISM hydrated-mineral subbenchmarks, namely, phyllosilicates, sulfates, and carbonates. The same training/validation/test partition and evaluation metrics are used for all methods. The quantitative results are summarized in Table IV.

As shown in Table IV, VMSA achieves the best performance on all three phyllosilicate, sulfate, and carbonate subbenchmarks, indicating that its performance advantage is not limited to a single mineral group or dependent on a specific evaluation metric.

On the phyllosilicate subbenchmark, VMSA achieves an OA of 97.33%, an AA of 93.10%, and a Kappa of 96.50%, outperforming all competing baseline methods. Compared with the strongest baseline, VMSA improves OA, AA, and Kappa by 10.55, 21.34, and 17.22 percentage points, respectively. This subbenchmark contains multiple spectrally similar classes with subtle local absorption differences and background mixing interference. The performance improvement on this subset therefore demonstrates the effectiveness of the proposed model in fine-grained discrimination of spectrally similar hydrated-mineral classes. Compared with models that mainly rely on local spectral–spatial patterns, VMSA not only improves the overall classification accuracy but also enhances the class-balanced recognition capability, indicating that the combination of explicit diagnostic priors and unified spectral–spatial contextual modeling provides substantial benefits for the discrimination of hard-to-classify mineral classes.

On the sulfate subbenchmark, VMSA reaches an OA of 99.41%, an AA of 98.51%, and a Kappa of 98.92%, maintaining a substantial performance lead over the competing methods. Compared with the strongest baseline, VMSA improves OA, AA, and Kappa by 19.99, 23.92, and 33.30 percentage points, respectively. Sulfate species often share partially overlapping diagnostic absorption intervals, and their spectral characteristics are jointly affected by hydration state, local spectral-shape fluctuations, and geological background conditions, which may lead to interclass confusion. The experimental results show that, for this mineral group that is sensitive to the structural organization of diagnostic absorption intervals, VMSA can more effectively reduce confusion among spectrally similar sulfate classes by explicitly modeling interval-level priors and jointly exploiting contextual spectral–spatial features.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
QUANTITATIVE COMPARISON OF THREE STANDARDIZED CRISM HYDRATED-MINERAL SUBBENCHMARKS BY DIFFERENT METHODS (%)

Method	Phyllo OA	Phyllo AA	Phyllo Kappa	Sulfate OA	Sulfate AA	Sulfate Kappa	Carbonate OA	Carbonate AA	Carbonate Kappa
A²S²K-ResNet [55]	85.13	68.18	76.73	79.42	74.59	65.62	82.79	80.56	60.88
SpectralFormer [30]	69.94	59.96	55.03	61.51	63.62	43.46	88.05	87.82	44.34
MCTGCL [32]	86.78	71.76	79.28	73.82	65.90	56.97	84.53	86.83	68.31
MSSFNet [31]	73.09	56.62	58.63	62.04	63.28	42.78	76.85	82.97	26.39
SSFTT [56]	85.78	69.49	77.28	71.20	71.59	55.42	92.50	91.73	58.35
MambaHSI [57]	79.00	66.47	67.12	74.77	68.21	58.14	86.56	89.21	42.34
VMSA	97.33	93.10	96.50	99.41	98.51	98.92	99.13	94.07	92.76

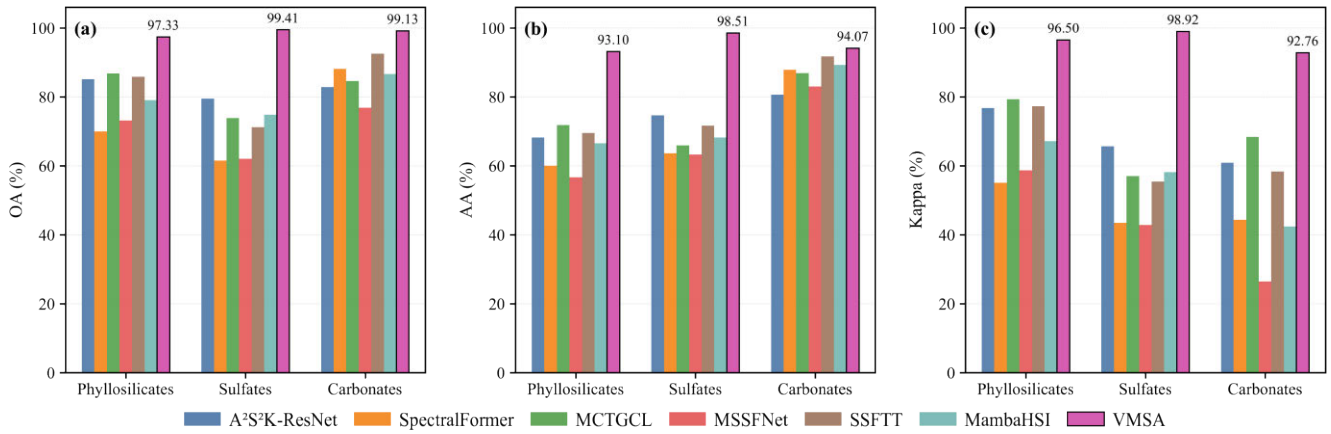


Fig. 8. Quantitative comparison of different methods on three standardized CRISM hydrated-mineral subbenchmarks. (a) OA. (b) AA. (c) Kappa coefficient.

The carbonate subbenchmark has fewer classes, but it is still sensitive to class imbalance and evaluation-metric instability; over-reliance on OA or a single metric can mask the actual classification stability. VMSA maintains the best performance on all three metrics, with an OA of 99.13%, an AA of 94.07%, and a Kappa of 92.76%. Compared with the strongest baseline for each metric, VMSA improves OA, AA, and Kappa by 6.63, 2.34, and 24.45 percentage points, respectively. These results indicate that its advantage is not an accidental gain on a single evaluation metric, but is reflected in stronger overall decision consistency and more stable interclass discrimination. Under the few-class condition, some models may also achieve relatively high OA by exploiting overall albedo differences or scene-specific background cues. However, VMSA still maintains clear advantages in AA and Kappa, suggesting that its classification decisions benefit from the joint use of diagnostic absorption information and contextual spectral–spatial representation, rather than being dominated only by scene-dependent brightness or local background patterns.

Collectively, the results from the three subbenchmarks show that the advantage of VMSA does not come from merely improving the OA on easy-to-classify classes. Instead, it is reflected in better class balance and stronger discrimination for hard-to-classify classes, minority classes, and mineral pairs with overlapping local absorption features. The performance differences across the three subbenchmarks are not solely caused by the number of scenes, but are jointly related to class composition, spectral similarity, interclass imbalance, and background mixing intensity. Overall, the results under the unified benchmark demonstrate that the performance gain of VMSA comes from its structural design and physically meaningful feature organization strategy, rather than from scene-specific or metric-specific advantages. A comprehensive comparison of the overall performance of different methods across the three standardized subbenchmarks is further presented in Fig. 8.

C. Lightweight and Efficiency Analysis

Beyond classification accuracy, we further quantify the parameter scale, computational complexity, training time, and inference efficiency of different methods under the unified benchmark setting. The results are summarized in Table V. To improve comparability, all complexity-related statistics are measured under a consistent input size and evaluation protocol, thereby minimizing bias caused by differences in implementation settings. Specifically, the model-complexity metrics are computed from the instantiated network architectures, the inference time is measured on the same GPU platform after warm-up through repeated forward passes, and the training time is reported as the wall-clock cost of one representative full training run.

From the overall results, A^{2}S^{2}K-ResNet maintains favorable compactness in both parameter count and inference latency,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

TABLE V
MODEL COMPLEXITY AND COMPUTATIONAL EFFICIENCY OF DIFFERENT METHODS ON THE UNIFIED HYDRATED-MINERAL BENCHMARK

Method	Params (M)	MACs (G)	FLOPs (G)	Training Time (s)	Inference Time (ms / 1000 patches)
A^2S^2K-ResNet	0.105	0.035	0.070	1550.25	869.76
SpectralFormer	0.121	0.004	0.008	559.59	1246.54
MCTGCL	0.238	0.028	0.057	4729.94	1087.04
MSSFNet	7.062	0.011	0.022	255.09	2391.59
SSFTT	0.153	0.011	0.023	196.87	508.94
MambaHSI	0.120	0.002	0.001	1991.91	1495.05
VMSA	0.271	0.024	0.048	1262.45	2271.26

indicating that convolution-based local spectral–spatial modeling still retains strong engineering efficiency. SpectralFormer and MambaHSI also have relatively small parameter scales, but their inference latency is not lower than that of A^2S^2K-ResNet, suggesting that a lighter parameter count does not necessarily translate into lower actual runtime cost. MCTGCL shows strong representational capability, but its training time is substantially higher. MSSFNet has the largest parameter count, implying that its multibranch fusion design comes with a relatively high model-scale cost. In contrast, SSFTT performs favorably in inference speed. Compared with these representative methods, VMSA uses only 0.271 M trainable parameters and 0.024 G MACs, and its full training time is 1262.45 s, while achieving the best overall recognition performance on the unified benchmark. Although its inference latency is not the lowest among all compared methods, the proposed model maintains a relatively compact architecture and low computational complexity, leading to a favorable balance between recognition accuracy and computational cost.

Therefore, the term lightweight in this study does not mean that VMSA is absolutely optimal in every single efficiency-related metric. Instead, it means that the proposed framework achieves a more favorable accuracy-efficiency tradeoff than existing representative methods under a relatively small parameter scale and modest computational complexity. For large-area hydrated-mineral screening and subsequent regional mapping tasks based on CRISM data, such a balance is more practically meaningful than pursuing either the lowest latency alone or the highest accuracy alone.

D. Ablation Experiment

To evaluate the individual contributions of the key components in VMSA, we conduct systematic ablation experiments on the unified phyllosilicate, sulfate, and carbonate benchmarks. Under identical training settings, a step-by-step module removal strategy is adopted to examine the effects of the diagnostic-prior pathway, the PGF module, the SAM, and the MCM on the final classification performance. The corresponding results are reported in Table VI.

The complete VMSA achieves the strongest overall performance across the three unified subbenchmarks, indicating that the individual components are not mutually replaceable, but provide complementary contributions at different levels. This demonstrates that the performance gain of VMSA does not arise from an incidental improvement of a single module, but from the joint effect of explicit diagnostic-prior modeling, PGF-based feature coupling, spectral–spatial attention, and long-range contextual relationship modeling.

The most evident performance degradation is observed when ablating the diagnostic-prior pathway, particularly on the phyllosilicate and sulfate benchmarks, indicating that contextual spectral–spatial feature learning alone is insufficient to support stable discrimination between hydrated-mineral classes with highly similar spectral shapes. Since PGF is designed for contextual-prior feature fusion, removing the DPB simultaneously eliminates the prior input to the fusion module. Therefore, this variant is denoted as w/o DPB/PGF and represents the model without the explicit diagnostic-prior pathway. Under this setting, the OA values on the phyllosilicate, sulfate, and carbonate benchmarks decrease to 91.25%, 92.93%, and 98.83%, respectively, while Kappa decreases to 88.54%, 87.33%, and 90.87%. These results show that continuum-removed diagnostic interval modeling, sample-adaptive interval responses, and class-conditional interval modulation provide VMSA with a critical discriminative basis beyond purely data-driven spectral–spatial modeling.

Ablating the PGF module also causes clear performance degradation across the three subbenchmarks, showing that simply preserving both the contextual and prior branches is insufficient; effective coupling between the two is equally important. Compared with the complete model, the w/o PGF variant reduces OA to 92.31%, 95.04%, and 98.53% on the phyllosilicate, sulfate, and carbonate benchmarks, respectively, with corresponding Kappa values dropping to 89.94%, 91.12%, and 87.79%. This result indicates that the advantage of VMSA does not come merely from adding an extra DPB, but from organizing contextual spectral–spatial representations and diagnostic absorption priors into a shared discriminative space via PGF, thereby enhancing mineral-sensitive absorption discrimination while preserving spatial-context information.

Ablating the SAM shows a relatively moderate and group-dependent influence on the classification performance. On the sulfate benchmark, the w/o SAM variant remains close to the complete model, with OA, AA, and Kappa changing from 99.41%, 98.51%, and 98.92% to 99.31%, 98.10%, and 98.75%, respectively. On the carbonate benchmark, removing SAM reduces AA and Kappa from 94.07% and 92.76% to 92.28% and 91.08%, respectively. On the phyllosilicate benchmark, the w/o SAM variant produces slightly higher values than the complete model. These results suggest that SAM is not the sole source of performance improvement, but acts as a contextual refinement component that strengthens spectral–spatial token interactions. Its contribution is therefore more closely related to improving feature organization and class-specific stability than to producing a uniform gain on every metric and mineral group.

Removing the MCM leads to a moderate performance decrease, especially on the phyllosilicate and sulfate benchmarks. Specifically, the w/o MCM variant reduces OA, AA, and Kappa to 95.41%, 91.03%, and 94.08% on the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
ABLATION RESULTS OF VMSA ON THE UNIFIED PHYLLOSILICATE, SULFATE, AND CARBONATE BENCHMARKS (OA/AA/KAPPA, %)

Variant	DPB	PGF	SAM	MCM	Phyllosilicates			Sulfates			Carbonates		
					OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
Final VMSA	✓	✓	✓	✓	97.33	93.10	96.50	99.41	98.51	98.92	99.13	94.07	92.76
w/o DPB	×	×	✓	✓	91.25	88.41	88.54	92.93	89.62	87.33	98.83	96.10	90.87
w/o PGF	✓	×	✓	✓	92.31	89.17	89.94	95.04	93.52	91.12	98.53	91.87	87.79
w/o SAM	✓	✓	×	✓	97.91	93.91	97.27	99.31	98.10	98.75	98.96	92.28	91.08
w/o MCM	✓	✓	✓	×	95.41	91.03	94.08	97.64	96.52	95.37	98.96	92.98	93.65

phyllosilicate benchmark, and to 97.64%, 96.52%, and 95.37% on the sulfate benchmark. These results indicate that the bidirectional Mamba context pathway contributes to long-range contextual consistency and improves the stability of fine-grained mineral discrimination. On the carbonate benchmark, the performance change is relatively small and metric-dependent, with OA and AA slightly decreasing but Kappa showing a marginal increase. This suggests that MCM contributes more clearly to mineral groups with more complex class composition and stronger spectral–spatial ambiguity, rather than being uniformly dominant across all benchmark groups.

Overall, the results in Table VI confirm that the performance gain of VMSA stems from a collaborative structure with clearly differentiated functional roles: DPB explicitly introduces mineral-sensitive absorption priors; PGF organizes the joint discrimination of contextual and prior-guided features; SAM enhances spectral–spatial token interaction; and MCM complements long-range structured contextual information. This module contribution pattern also supports the subsequent hard-pair analysis and interval-level spectral verification, confirming that VMSA's performance improvement is not a random gain limited to specific scenes or individual classes, but is tied to its overall structural design and physically meaningful feature organization.

E. Cross-Scene Mapping Comparison on Representative CRISM Scenes

Beyond the quantitative comparison on the unified benchmark, we further conduct a qualitative comparison of the spatial mapping results from different methods on six representative CRISM targeted observations. This analysis provides complementary visual evidence for assessing spatial continuity, local boundary preservation, and recognition stability for weak-absorption or sparsely distributed mineral classes in real Martian geological settings. Figs. 9–11 present the pixel-level classification results in three representative regions: Columbus Crater, Mawrth Vallis, and Nili Fossae, respectively. Each figure simultaneously provides the ground truth, the prediction results of VMSA, and the mapping outputs of several representative baseline methods, enabling a direct comparison of the differences between different methods in spatial structure recovery and local misclassification patterns.

The Columbus Crater scene shown in Fig. 9 demonstrates that VMSA outperforms the compared methods in terms of regional integrity and boundary continuity of the major mineral units, with fewer isolated misclassified pixels, especially in sparsely distributed areas and near lithological contact boundaries.

TABLE VII
EXTERNAL TRANSFERABILITY COMPARISON ON THE PAVIA UNIVERSITY DATASET (%)

Method	OA	AA	Macro-F1	Kappa
A²S²K-ResNet	99.79	99.61	99.62	99.72
SpectralFormer	97.40	96.45	95.98	96.55
MambaHSI	99.46	99.03	99.08	99.29
VMSA	99.66	99.49	99.42	99.55

The Mawrth Vallis scene presented in Fig. 10 further shows that VMSA maintains favorable spatial compactness in regions with small patch sizes and highly fragmented spatial distribution, while most baseline methods are more prone to background misclassification and patchy fragmentation.

The Nili Fossae scene shown in Fig. 11 illustrates that VMSA achieves more stable preservation of elongated mineral patches and local boundary regions for spectrally similar classes with sparse spatial distribution, delivering more stable spatial discrimination results in complex geological backgrounds.

Overall, the spatial mapping patterns presented in Figs. 9–11 are broadly consistent with the quantitative results on the unified benchmark presented in Sections IV-B–IV-D. The advantage of VMSA is not limited to improving pixel-level accuracy, but is also reflected in more continuous mineral distributions, fewer isolated misclassified pixels, and more stable preservation of local spatial structures in challenging regions. These scene-level results thus serve as complementary visual evidence beyond the benchmark-level quantitative evaluation, further validating the spatial discrimination stability of the proposed method in real Martian geological backgrounds.

F. External Transferability Evaluation on Public Hyperspectral Benchmarks

To evaluate the transferability of VMSA beyond CRISM hydrated-mineral mapping, we conduct experiments on three public hyperspectral benchmarks: Pavia University, Indian Pines, and Salinas. These datasets represent standard terrestrial HSI classification tasks that are independent of Martian

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

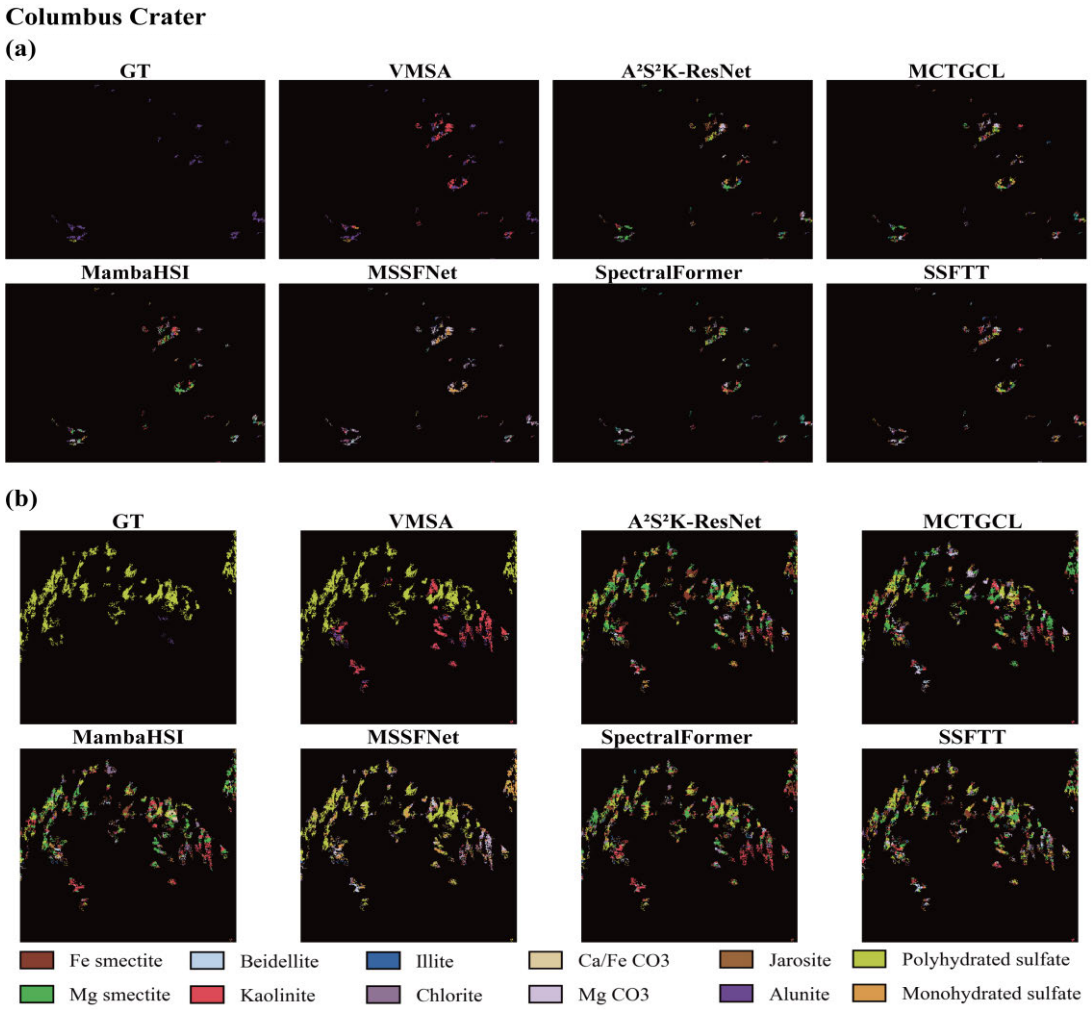


Fig. 9. Qualitative comparison of mineral classification maps on representative CRISM scenes from Columbus Crater. (a) FRT00013EEF. (b) FRT0000750A.

TABLE VIII
EXTERNAL TRANSFERABILITY COMPARISON ON THE INDIAN PINES DATASET (%)

Method	OA	AA	Macro-F1	Kappa
A²S²K-ResNet	71.20	91.11	81.12	68.47
SpectralFormer	89.97	86.38	87.57	88.60
MambaHSI	96.05	94.27	94.28	95.50
VMSA	96.48	97.99	95.61	95.99

TABLE IX
EXTERNAL TRANSFERABILITY COMPARISON ON THE SALINAS DATASET (%)

Method	OA	AA	Macro-F1	Kappa
A²S²K-ResNet	99.93	99.91	99.89	99.92
SpectralFormer	98.20	99.00	98.92	98.00
MambaHSI	99.35	99.63	99.59	99.28
VMSA	99.69	99.71	99.72	99.65

mineral labels and CRISM preprocessing. For this evaluation, VMSA is used in a generic hyperspectral classification setting, without the CRISM-specific diagnostic intervals used for hydrated-mineral analysis. Therefore, the experiments examine the general spectral–spatial representation ability of VMSA rather than its mineral-specific prior modeling. All methods, including A^{2}S^{2}K-ResNet, SpectralFormer, MambaHSI, and VMSA, are evaluated under the same splits, input settings, training protocol, and metrics. OA, AA, Macro-$F1$, and Kappa are reported in Tables VII–IX, with the corresponding classification maps shown in Figs. 12–14.

As shown in Table VII, VMSA achieves competitive performance on the Pavia University dataset, with an OA of 99.66%, an AA of 99.49%, a Macro-$F1$ of 99.42%, and a Kappa of 99.55%. These results are slightly lower than those of A^{2}S^{2}K-ResNet, which obtains the best performance on this dataset, but remain higher than those of SpectralFormer and MambaHSI. The classification maps in Fig. 12 show that VMSA preserves the main spatial structures of urban land-cover classes and produces spatially coherent predictions in most regions. Compared with SpectralFormer and MambaHSI, the VMSA map shows fewer local noisy predictions in several fragmented areas, although A^{2}S^{2}K-ResNet remains slightly

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

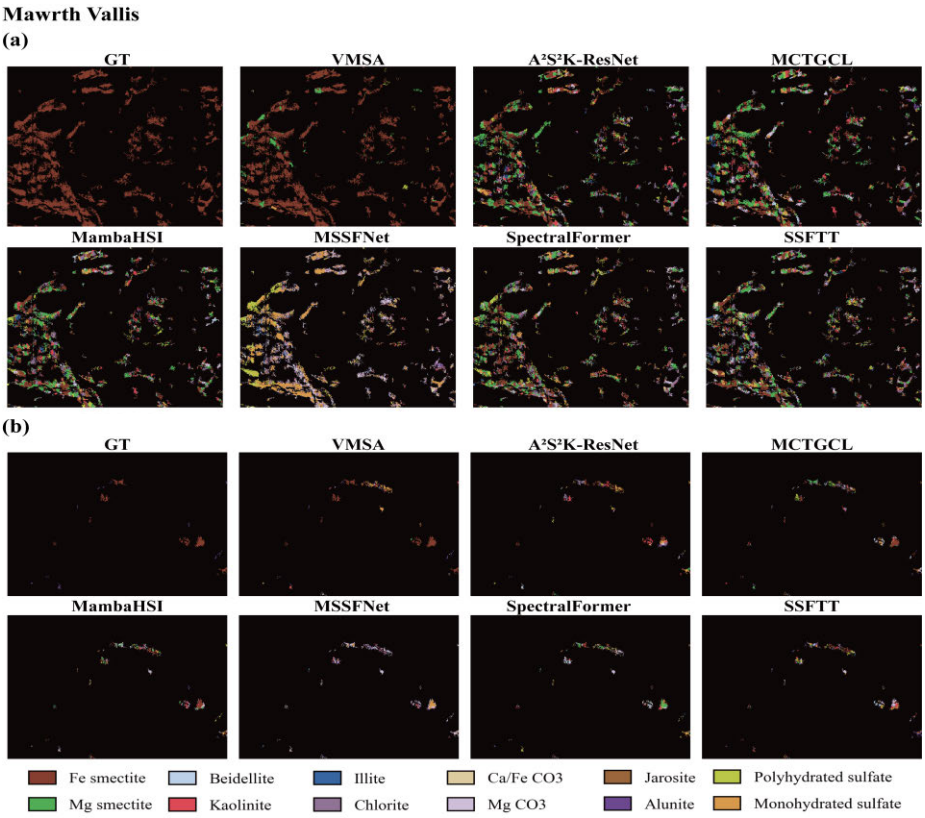


Fig. 10. Qualitative comparison of mineral classification maps on representative CRISM scenes from Mawrth Vallis. (a) FRT00009326. (b) FRT0000C467.

stronger in the quantitative metrics. This result indicates that the spectral–spatial representation capability of VMSA can effectively handle urban land-cover classification characterized by fine spatial structures and high interclass separability.

On the more challenging Indian Pines dataset, the results in Table VIII show that VMSA achieves the best overall performance among the compared methods, with an OA of 96.48%, an AA of 97.99%, a Macro-$F1$ of 95.61%, and a Kappa of 95.99%. Compared with MambaHSI, VMSA improves OA, AA, Macro-$F1$, and Kappa by 0.43%, 3.72%, 1.33%, and 0.49%, respectively. The improvement in AA is particularly notable, suggesting that the proposed spectral–spatial representation strategy is effective for class-balanced recognition under more difficult class distributions and limited training samples. The classification maps in Fig. 13 further support this observation: VMSA better preserves field-level spatial continuity and reduces scattered misclassifications compared with several baseline maps. This visual behavior is consistent with its higher AA and Macro-$F1$, indicating improved recognition stability for minority or spectrally similar crop classes.

For the Salinas dataset, the results in Table IX show that all compared methods achieve high classification accuracy, reflecting the relatively strong spectral separability of this benchmark. A^2S^2K-ResNet obtains the best overall performance, whereas VMSA remains highly competitive, achieving an OA of 99.69%, an AA of 99.71%, a Macro-$F1$ of 99.72%, and a Kappa of 99.65%. Compared with SpectralFormer and MambaHSI, VMSA achieves higher values on all four metrics, although it does not exceed A^2S^2K-ResNet on this dataset. The corresponding classification maps in Fig. 14 show that VMSA produces compact and continuous predictions for large agricultural parcels, with clear boundaries between major land-cover categories. Since most methods already perform well on this dataset, the visual differences are relatively subtle, which is consistent with the saturated quantitative performance in Table IX.

Overall, the public HSI experiments show that VMSA has external transferability beyond the CRISM hydrated-mineral benchmark. At the same time, these results should be interpreted as complementary evidence for the generic spectral–spatial modeling capability of the proposed framework, rather than as direct validation of CRISM-specific mineral diagnostic priors. The main mineral-mapping conclusions of this study remain based on the standardized CRISM hydrated-mineral subbenchmarks, the qualitative cross-scene CRISM mapping comparison, and the image-extracted spectral verification. Together, the public HSI experiments provide additional evidence that the proposed spectral–spatial modeling strategy is not narrowly tuned to a single planetary dataset.

G. Simulation Analysis for Spectrally Similar Mineral Classes

To further examine the behavior of VMSA under controlled spectral ambiguity, we conduct a mechanism-level stress test on representative spectrally similar mineral pairs. This experiment is designed to complement the real CRISM benchmark by isolating model sensitivity to controlled perturbations, rather than serving as another standard classification benchmark. Therefore, the controlled comparison focuses on Full VMSA and its w/o DPB/PGF counterpart,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

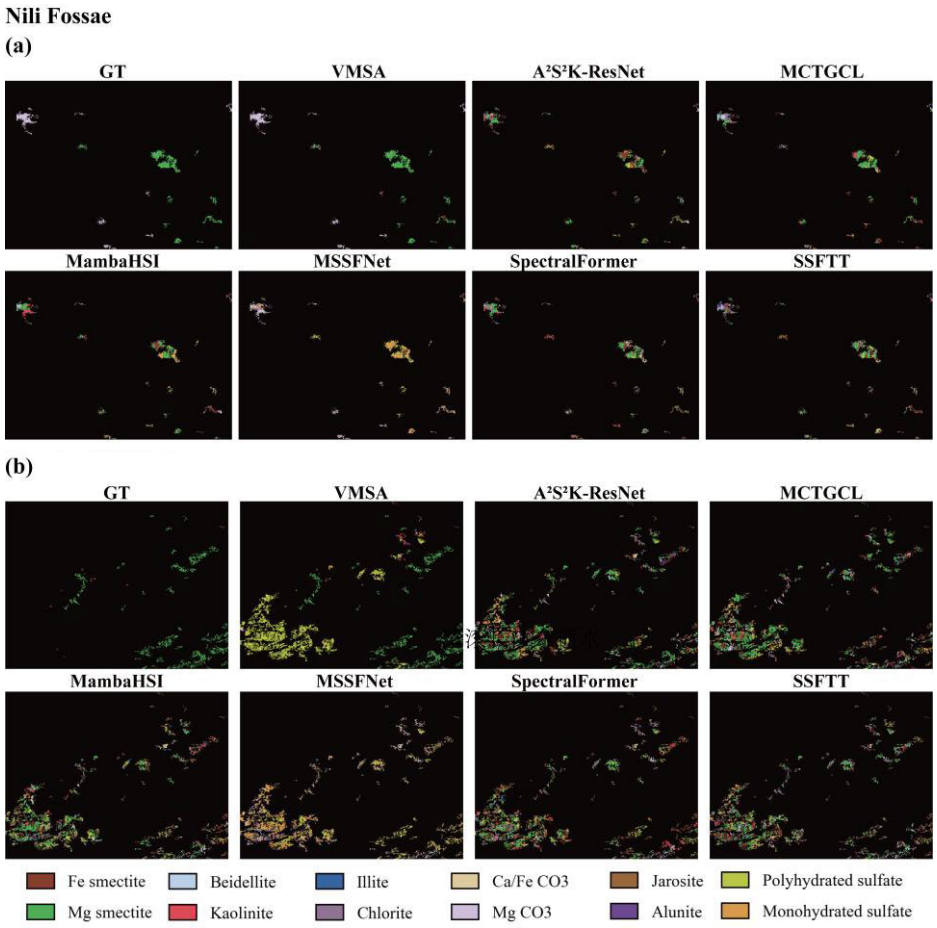


Fig. 11. Qualitative comparison of mineral classification maps on representative CRISM scenes from Nili Fossae. (a) FRT0000A09C. (b) FRT00009971.

for which the same endpoint-mined perturbation paths can be applied with and without the diagnostic-prior pathway. Three hard pairs are considered, including Fe-smectite versus Mg-smectite, jarosite versus alunite, and $MgCO_3$ versus $CaCO_3/FeCO_3$. These pairs represent typical fine-grained discrimination problems in phyllosilicates, sulfates, and carbonates, respectively.

For each pair, endpoint-mined validation samples, representing high-confidence target and competing samples, are used to construct controlled perturbation paths. The perturbation coefficient α decreases from 1.0 to 0.5, where $\alpha = 1.0$ represents the clean target endpoint and smaller values indicate stronger perturbation toward the competing condition. Two perturbation modes are evaluated. In the background/context perturbation, the diagnostic-prior input is preserved while the contextual spectral–spatial input is perturbed. In the diagnostic-prior perturbation, the contextual spectral–spatial input is preserved while the continuum-removed diagnostic-prior input is perturbed. The target-minus-competing probability margin, P(target) − P(competing), is used to quantify the stability of hard-pair discrimination, together with the correct-target rate at the strongest perturbation level.

As shown in Fig. 15, Full VMSA maintains larger target-minus-competing margins than the w/o DPB/PGF variant under background/context perturbation. At $\alpha = 0.5$, Full VMSA retains positive margins of 0.543, 0.864, and 1.000 for Fe-smectite versus Mg-smectite, jarosite versus alunite, and $MgCO_3$ versus $CaCO_3/FeCO_3$, respectively. The corresponding correct-target rates are 0.750, 0.859, and 1.000. In contrast, the w/o DPB/PGF variant drops to zero margin for all three hard pairs under the same perturbation level, with correct-target rates of 0.453, 0.469, and 0.500, respectively. These results indicate that the diagnostic-prior pathway and PGF help to stabilize hard-pair discrimination when background or contextual information becomes ambiguous.

The Fe-smectite versus Mg-smectite pair remains the most challenging case, as Full VMSA shows a larger margin decrease under strong background/context perturbation than the other two pairs. This is consistent with the subtle difference between Fe- and Mg-related metal–OH absorption structures. Nevertheless, Full VMSA still maintains a positive margin and a higher correct-target rate than the w/o DPB/PGF variant, suggesting that diagnostic-prior modeling provides useful support even under severe contextual perturbation.

Under diagnostic-prior perturbation, Full VMSA also maintains high margins at $\alpha = 0.5$, with values of 0.965, 0.895, and 0.977 for Fe-smectite versus Mg-smectite, jarosite versus alunite, and $MgCO_3$ versus $CaCO_3/FeCO_3$, respectively. This suggests that the final model does not rely only on a single isolated diagnostic cue but also combines diagnostic-prior information with contextual spectral–spatial representation. The w/o DPB/PGF variant is largely insensitive to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

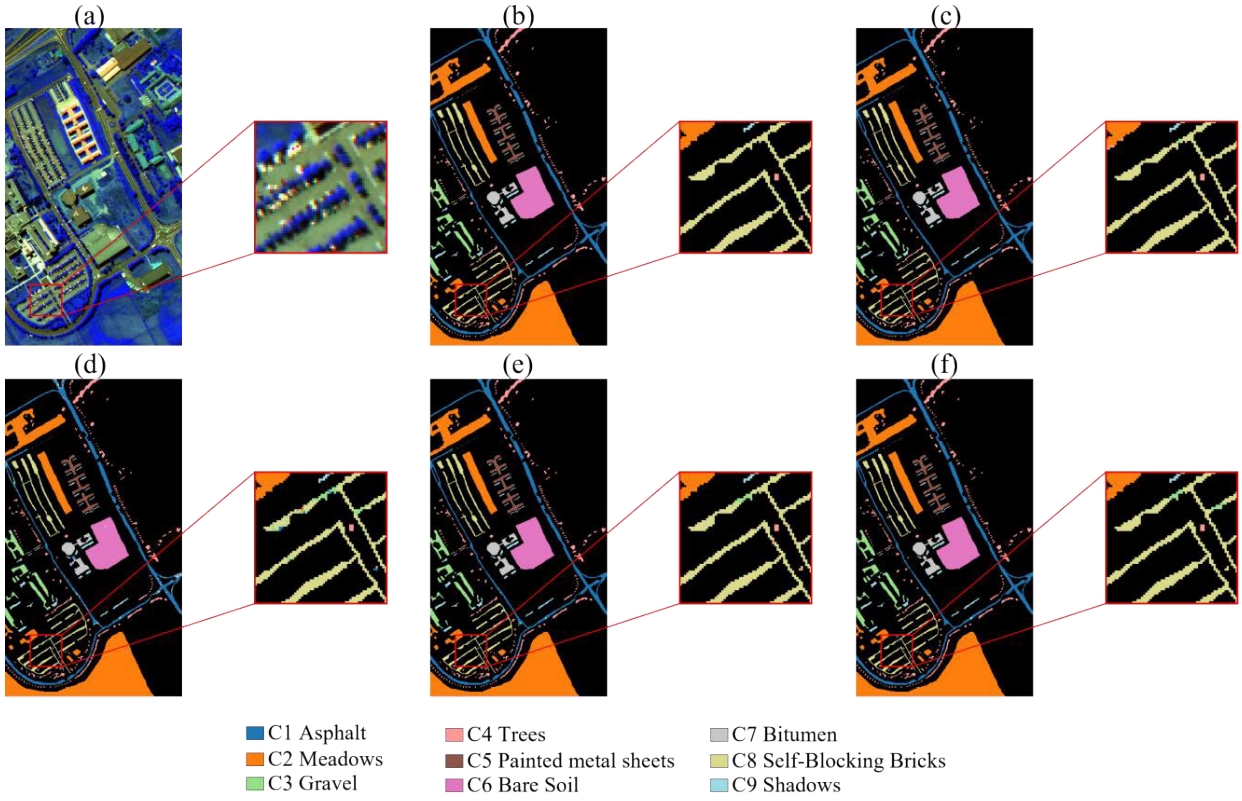


Fig. 12. Classification maps on the Pavia University dataset. (a) False-color composite. (b) Ground truth. (c) A^{2}S^{2}K-ResNet. (d) SpectralFormer. (e) MambaHSI. (f) VMSA. The class-color legend is shown below each map, and the detailed mapping between class codes and land-cover categories is provided in the supplementary class-name table.

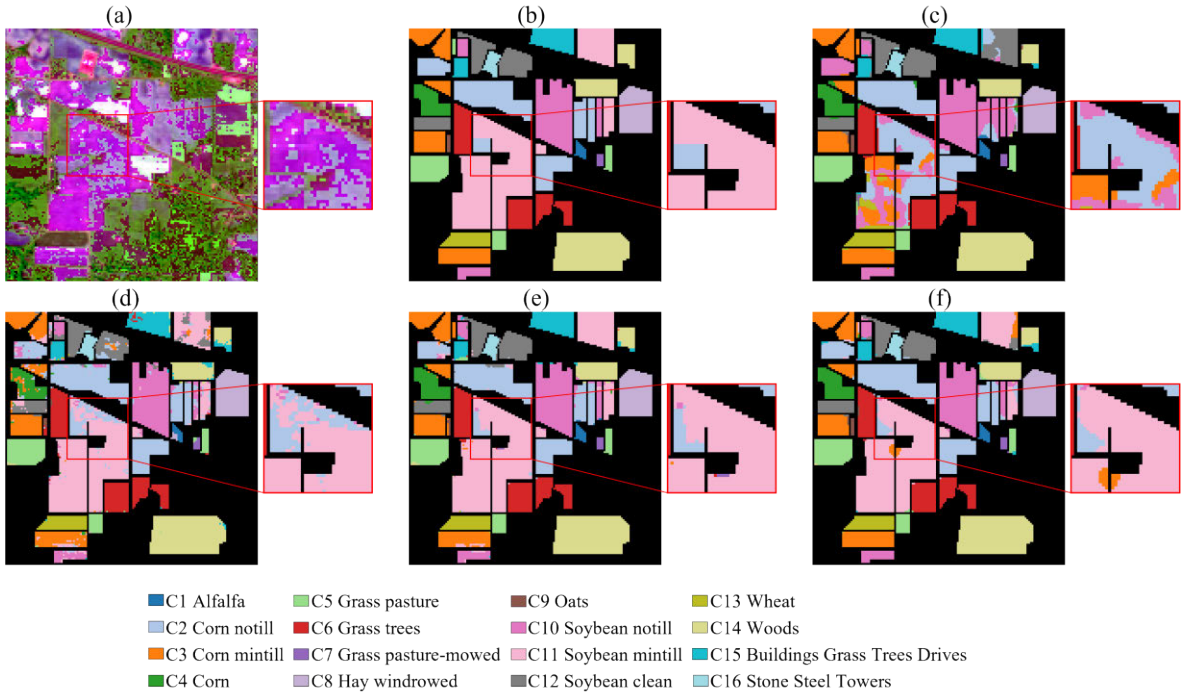


Fig. 13. Classification maps on the Indian Pines dataset. (a) False-color composite. (b) Ground truth. (c) A^{2}S^{2}K-ResNet. (d) SpectralFormer. (e) MambaHSI. (f) VMSA. The class-color legend is shown below each map, and the detailed mapping between class codes and land-cover categories is provided in the supplementary class-name table.

diagnostic-prior perturbation because the diagnostic-prior pathway is removed in this variant; therefore, its stable curve under this perturbation mode should not be interpreted as stronger mineral-prior robustness.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

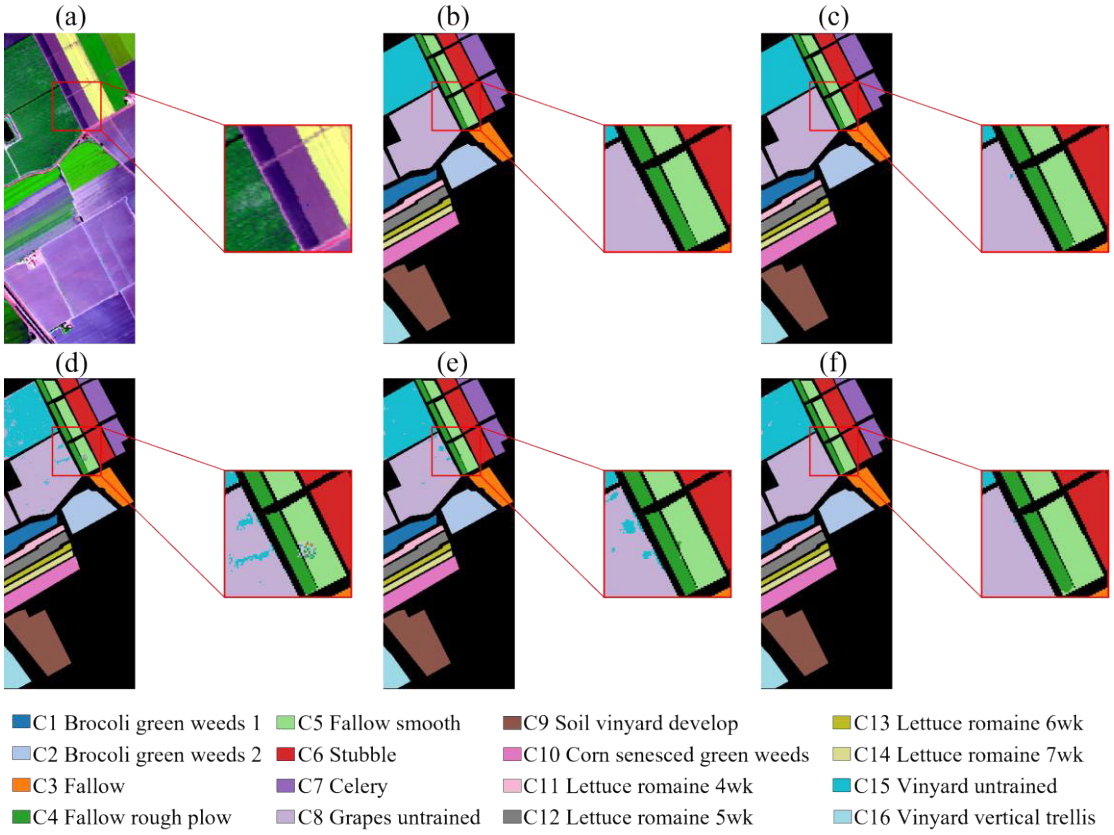


Fig. 14. Classification maps on the Salinas dataset. (a) False-color composite. (b) Ground truth. (c) A^2S^2K-ResNet. (d) SpectralFormer. (e) MambaHSI. (f) VMSA. The class-color legend is shown below each map, and the detailed mapping between class codes and land-cover categories is provided in the supplementary class-name table.

Overall, the controlled simulation results provide mechanism-level evidence that VMSA maintains more stable discrimination for representative spectrally similar mineral pairs, especially when background/context information becomes ambiguous. This analysis is used to examine model behavior under simplified and controllable perturbation conditions. It does not replace real CRISM spectral verification; the mineralogical plausibility of predicted classes is further examined using image-extracted spectra in Section V-B.

H. Cross-Scene Mapping Consistency With Spectral Parameter Composites

To further validate the regional mapping performance of VMSA in real complex geological scenarios, we apply the fully trained model to three CRISM targeted observations in the Mawrth Vallis region (FRT0001454D, FRT0000A600, and FRT00016E2D), which provide representative hydrated-mineral-bearing application scenes for cross-scene evaluation. The resulting pixel-level mineral classification maps are compared with the corresponding CRISM spectral summary products (SSPs), which are widely used for mineral screening in Martian hyperspectral analysis [12], [13]. Fig. 16 presents the VMSA mapping results and their matching SSP composites for the three scenes, to examine the spatial consistency between the model's predicted mineral distributions and standard mineral-sensitive spectral parameters. This cross-scene evaluation serves as complementary geological plausibility validation beyond the quantitative results on the unified benchmark, rather than a replacement for the core quantitative assessment.

As shown in Fig. 16, the main distribution of polyhydrated sulfate predicted by VMSA in the A600 and 16E2D scenes shows clear spatial correspondence with the high-response regions in the HYD composite maps, indicating that the sulfate-related units identified by the model do not deviate from the primary regions indicated by standard hydration-sensitive parameters. For Fe/Mg phyllosilicates, the results for the 1454D and 16E2D scenes demonstrate that the Mg-smectite units mapped by VMSA correspond closely to the high-response regions dominated by the D2300 parameter in the PFM composites; in the A600 scene, the Fe-smectite distribution predicted by VMSA also maintains good consistency with the relevant high-response regions in the PHY composite. Beyond spatial location, the patch morphology and local continuity of different mineral units in Fig. 16 are also broadly compatible with the regional geological structures revealed by the parameter composites.

It should be emphasized that the consistency shown in Fig. 16 is not definitive proof of mineralogical identification, but a map-level spatial consistency check. In other words, the outputs of VMSA do not deviate from the core

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

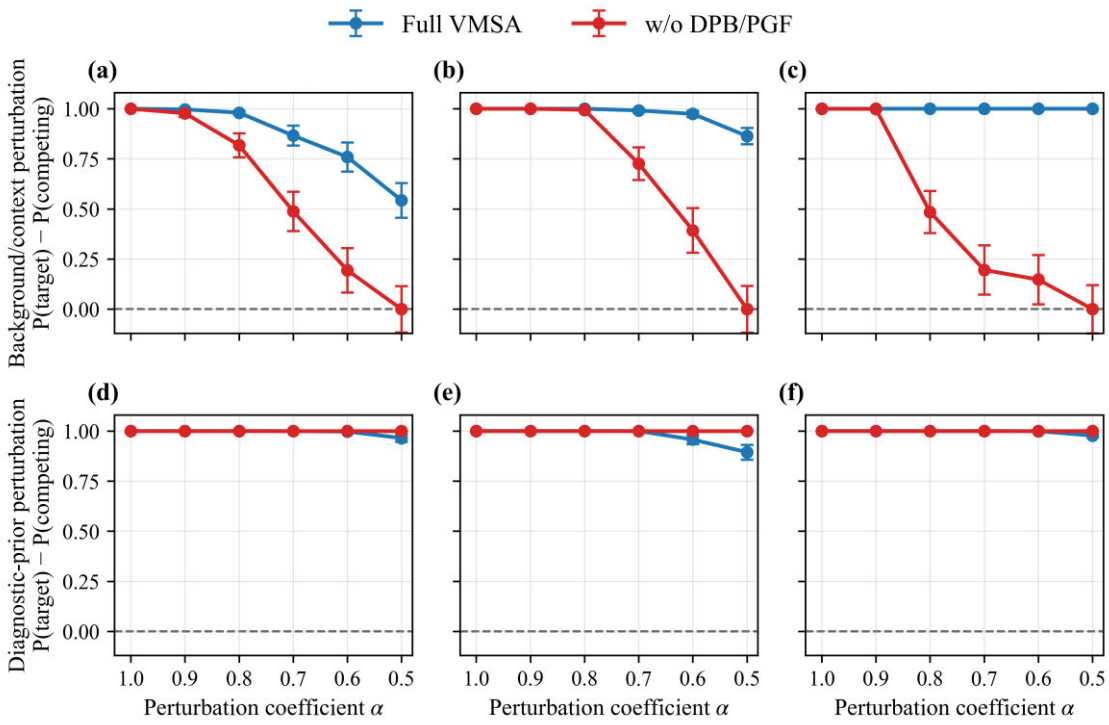


Fig. 15. Controlled mechanism stress test for representative spectrally similar mineral pairs. The curves show the mean target-minus-competing probability margin, P(target) − P(competing), as the perturbation coefficient α decreases from 1.0 to 0.5. The first and second rows correspond to background/context perturbation and diagnostic-prior perturbation, respectively. Error bars indicate the standard error of the mean, and the dashed line marks zero margin. (a) and (d) Fe-smectite versus Mg-smectite. (b) and (e) Jarosite versus alunite. (c) and (f) $MgCO_3$ versus $CaCO_3/FeCO_3$.

mineral-sensitive spatial patterns reflected by traditional CRISM parameter composites, but generate regionally coherent mapping results that align with standard parameter-based analysis in real geological scenarios. To further resolve spectral differences among the hard mineral pairs, Section V-B compares ratioed CRISM benchmark spectra and quantifies continuum-normalized band-depth contrasts at the selected diagnostic intervals.

V. Discussion

A. Hard-Mineral-Pair Discrimination and Diagnostic-Interval Preference

Although the overall metrics on the unified benchmark confirm that VMSA achieves strong performance across the phyllosilicate, sulfate, and carbonate subbenchmarks, the more critical finding is that this performance gain does not primarily stem from incremental improvements to easily distinguishable classes. Instead, it is driven by a significant reduction in misclassification between spectrally similar hydrated-mineral classes. Fig. 17 illustrates this behavior from three complementary perspectives: class confusion structure, class-conditional diagnostic-interval preference, and globally learned emphasis across diagnostic intervals.

The confusion matrix in Fig. 17(a) shows that the model's dominant misclassifications are not widely dispersed across unrelated classes, but mainly concentrated on genuinely challenging hard mineral pairs: Fe-smectite versus Mg-smectite, jarosite versus alunite, polyhydrated sulfate versus monohydrated sulfate, and $CaCO_3/FeCO_3$ versus $MgCO_3$. This pattern confirms that the primary discrimination pressure on the unified benchmark arises from mineral pairs with nearly overlapping diagnostic absorption structures, rather than coarse separation between background and target classes, or between spectrally distinct mineral groups. In other words, VMSA's performance improvement is not driven by scene-specific background variations or overall brightness differences, but by targeted suppression of misclassification between physically meaningful hard pairs.

Fig. 17(b) further demonstrates that different mineral classes exhibit distinct, class-conditional diagnostic-interval preferences, rather than sharing a uniform average spectral response. Fe/Mg phyllosilicate-related classes rely primarily on M–OH-sensitive intervals near 2.3 μm; sulfate classes depend jointly on hydration-sensitive intervals at 1.4–1.9 μm and local absorption differences in the 2.1–2.4-μm range; and carbonate-related classes focus on the coupled organization of paired absorption features near 2.3 and 2.5 μm. This behavior confirms that the learned interval preferences are not random attention redistribution, but broadly consistent with the diagnostic wavelength organization standardly used in CRISM spectral-parameter analysis and interpretable mineral mapping.

The globally learned emphasis across diagnostic intervals is presented in Fig. 17(c). Rather than assigning near-uniform weights to all intervals, the model selectively emphasizes several key intervals most critical for separating the hard mineral pairs. This indicates that the interval-level modeling introduced by the explicit DPB does not degenerate into

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

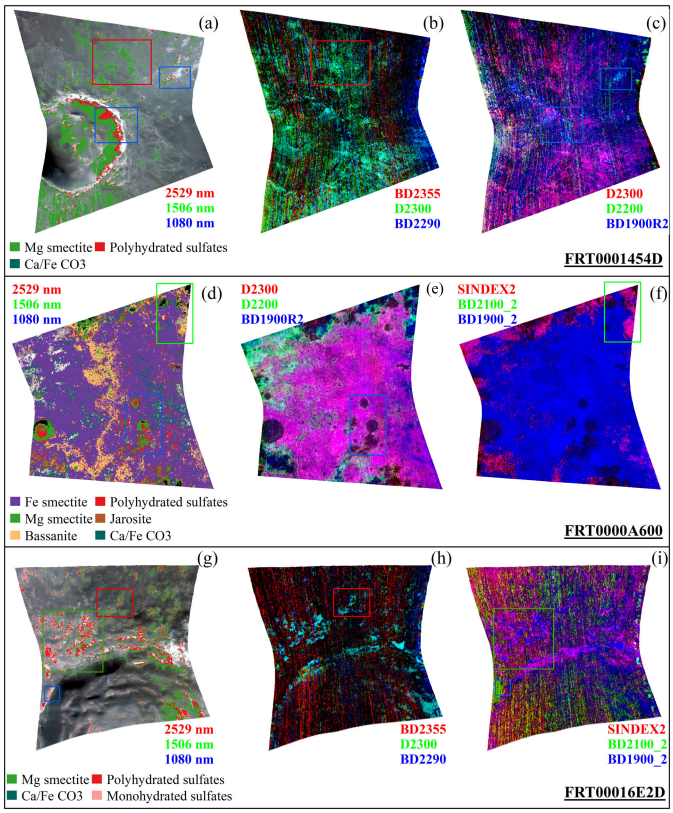


Fig. 16. Comparison between VMSA mineral mapping results and CRISM spectral-index RGB composites in the Mawrth Vallis region. Three hydrated-mineral-bearing CRISM targeted observations are shown: (a)–(c) FRT0001454D, (d)–(f) FRT0000A600, and (g)–(i) FRT00016E2D. (a), (d), and (g) Pixelwise mineral classification maps generated by the final VMSA model, while the corresponding panels show SSP RGB composites formed from selected mineral-sensitive spectral indices.

full-spectrum average processing, but progressively develops targeted emphasis on key diagnostic intervals during unified end-to-end training. When interpreted jointly, Fig. 17(a) and (b) make clear that VMSA's core performance gain is concentrated on the hard mineral pairs, and this hard-pair discrimination advantage is directly aligned with the model's class-dependent diagnostic-interval preferences.

It should be emphasized that the interval-preference patterns in Fig. 17 are not interpreted here as strict recovery of mineral templates, nor as evidence that the model has automatically completed full mineralogical reasoning. A more appropriate interpretation is that, under the unified benchmark, VMSA's decision-making process establishes a stable, physically meaningful correspondence with mineral-sensitive diagnostic intervals, confining misclassification to genuinely challenging classes rather than dispersing errors broadly across unrelated categories. For this study, this coordination between discriminative behavior and diagnostic absorption structure holds greater interpretive value than incremental gains in overall classification accuracy.

B. Pairwise Spectral Verification and Mineralogical Consistency

The class-level spectral context in Fig. 7 and the pairwise comparisons in Figs. 18 and 19 provide complementary evidence for interpreting the mineralogical plausibility of VMSA predictions. Fig. 7 compares ratioed CRISM benchmark spectra, MICA-corrected CRISM type-reference spectra, and laboratory reference spectra. Building on this class-level context, Figs. 18 and 19 focus on the most challenging mineral pairs identified from the confusion structure of the unified benchmark. For each class, the representative CRISM scene was selected according to the largest number of labeled pixels, and continuum normalization was applied to individual ratioed spectra before class-level aggregation. The resulting median spectra and interquartile envelopes characterize the central absorption morphology and within-class variability, whereas pairwise median band-depth differences highlight wavelength regions that contribute most directly to class separation.

The selected diagnostic centers indicate that discrimination is generally governed by coordinated differences across multiple absorption regions rather than by a single isolated feature. Across the six hard mineral pairs, 16 of the 18 prespecified diagnostic centers showed bootstrap confidence intervals for the pairwise band-depth contrast that did not cross zero. This result indicates that the observed spectral contrasts are stable across the retained labeled-pixel populations and are not driven by a small number of individual spectra. These contrast patterns provide a spectral basis for interpreting the class-dependent diagnostic-interval preferences shown in Fig. 17.

For Fe-smectite and Mg-smectite, the continuum-normalized spectra exhibit consistent separation near 1.42, 1.91, and 2.31 μm, with deeper absorption for Mg-smectite at all three retained centers. The contrast near 2.31 μm is particularly pronounced, consistent with the sensitivity of this spectral region to differences in octahedral cation composition and metal–OH absorption behavior. Jarosite and alunite show stable contrast at approximately 1.48, 1.98, and 2.18 μm, where alunite displays systematically deeper absorptions. These differences reflect the distinct hydroxyl- and sulfate-related absorption organization of the two minerals. The sulfate hydration pair also exhibits a clear multiband pattern: polyhydrated sulfate has stronger absorption near 1.43 μm, whereas monohydrated sulfate shows deeper bands near 2.14 and 2.40 μm. This complementary contrast is consistent with differences in hydration state and demonstrates that separation of the two sulfate classes depends on the joint organization of water-related and sulfate-related absorptions.

The carbonate pair shows a more localized but still stable contrast. $MgCO_3$ exhibits deeper band depths at the retained centers near 1.41 and 2.30 μm, whereas the center near 1.94 μm does not show a sufficiently stable pairwise contrast and is therefore not retained as a primary discriminative feature. This result suggests that discrimination between $CaCO_3$/$FeCO_3$ and $MgCO_3$ is associated primarily with the combined behavior of the selected carbonate-sensitive regions rather than with uniform differences across the full spectral range. For the phyllosilicate pairs, illite and chlorite display stable contrasts near 1.41, 2.21, and 2.35 μm, with deeper absorption for illite at the retained centers. These differences are consistent with the contrasting contributions of Al–OH and Mg–OH/Fe–OH bonding environments. Beidellite and kaolinite are separated most consistently near 1.41 and 2.21 μm,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

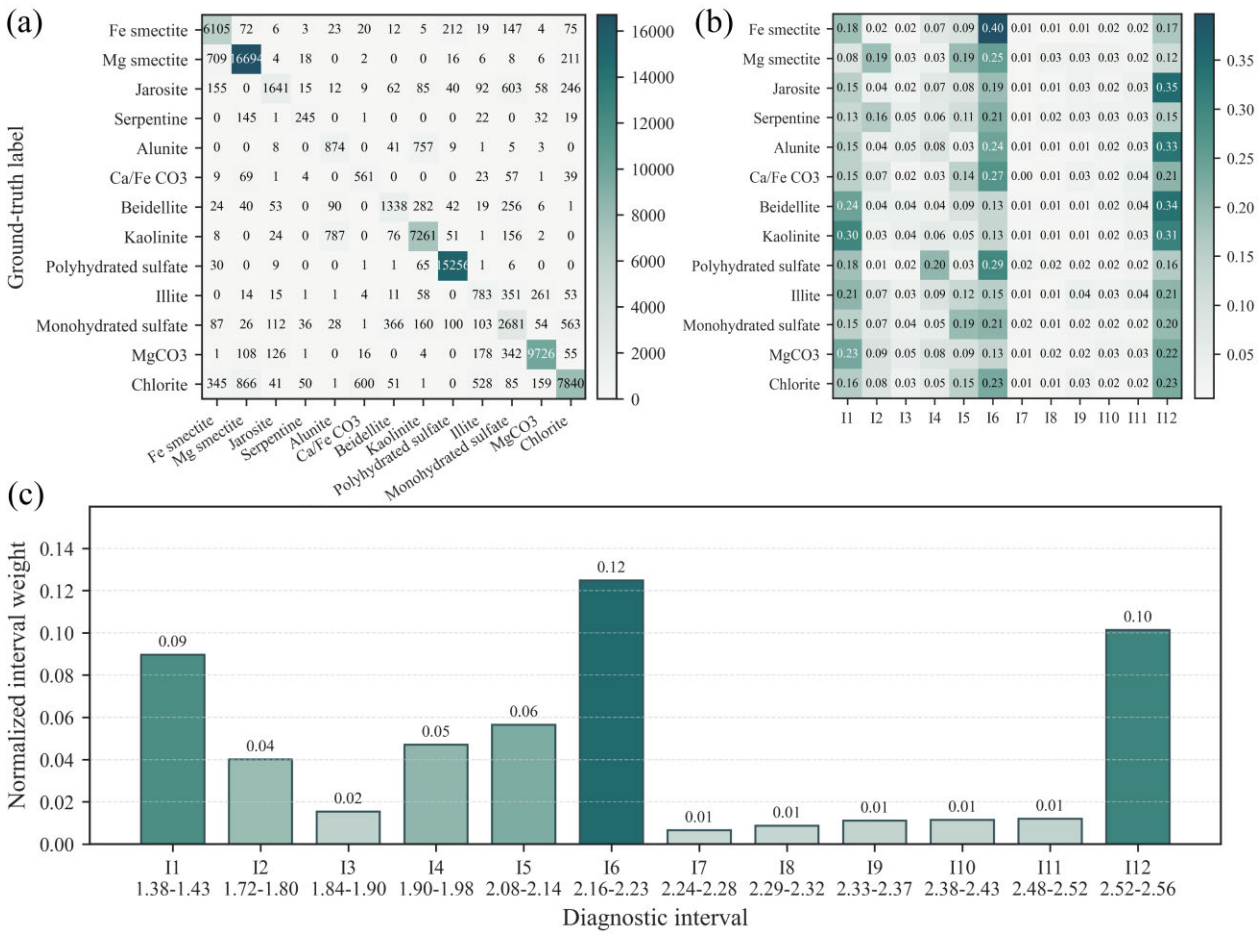


Fig. 17. Hard-mineral-pair discrimination and diagnostic-interval behavior on the unified hydrated-mineral benchmark. (a) Confusion matrix on the unified test set. (b) Class-conditional diagnostic-interval preference for representative hydrated-mineral classes. (c) Global learned emphasis over diagnostic intervals.

where kaolinite shows deeper absorption. Although a narrow feature near 1.65 μm is present in the beidellite spectra, it is not treated as a diagnostic basis for discrimination because it does not correspond to a prespecified mineral-sensitive center.

The hard-pair analyses indicate that VMSA does not distinguish spectrally similar hydrated minerals solely through broad spectral level, scene-dependent brightness, or spatial texture. Instead, the model's classification behavior is associated with reproducible differences in the relative depth and organization of diagnostic absorptions across multiple wavelength regions. This correspondence is particularly evident for the Fe-/Mg-smectite, jarosite–alunite, polyhydrated–monohydrated sulfate, and illite–chlorite pairs, for which all retained diagnostic centers show stable pairwise contrasts. The weaker or nonretained centers in the carbonate and beidellite–kaolinite comparisons further emphasize that discriminative information is concentrated in specific mineral-sensitive intervals rather than distributed uniformly across the spectrum.

C. Practical Scope and Limitations

The results of this study show that the VMSA improves discrimination between spectrally similar hydrated-mineral classes on the unified hydrated-mineral benchmark, while exhibiting favorable spatial continuity and mineralogical consistency in representative geological scenes. These results are best understood as an enhancement to automated classification and mapping workflows, rather than a replacement for traditional CRISM mineralogical analysis. For Martian orbital imaging spectroscopy, SSPs, ratioed spectra, CAT-assisted analysis, and MICA-type reference spectra remain essential tools for mineral identification and geological interpretation. The appropriate positioning of VMSA is therefore as a complementary component of these established workflows, not a standalone system for full mineralogical reasoning [13], [17], [20].

In terms of methodological design, the core value of VMSA lies in its integration of unified spectral–spatial contextual modeling and explicit diagnostic-absorption priors into a single classification framework. This design moves beyond models that rely primarily on local texture or overall brightness differences for class separation, enabling the establishment of more stable discrimination boundaries for hard mineral pairs. Combined with the preceding results, this performance gain does not stem from the incidental effect of a single module, but from the synergy among long-range contextual modeling via the MCM, spectral-attention modeling via the SAM, explicit mineralogical prior embedding via the DPB, and effective feature coupling via the PGF module. For CRISM automated recognition tasks, this structural integration strengthens the connection between benchmark-level classi-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

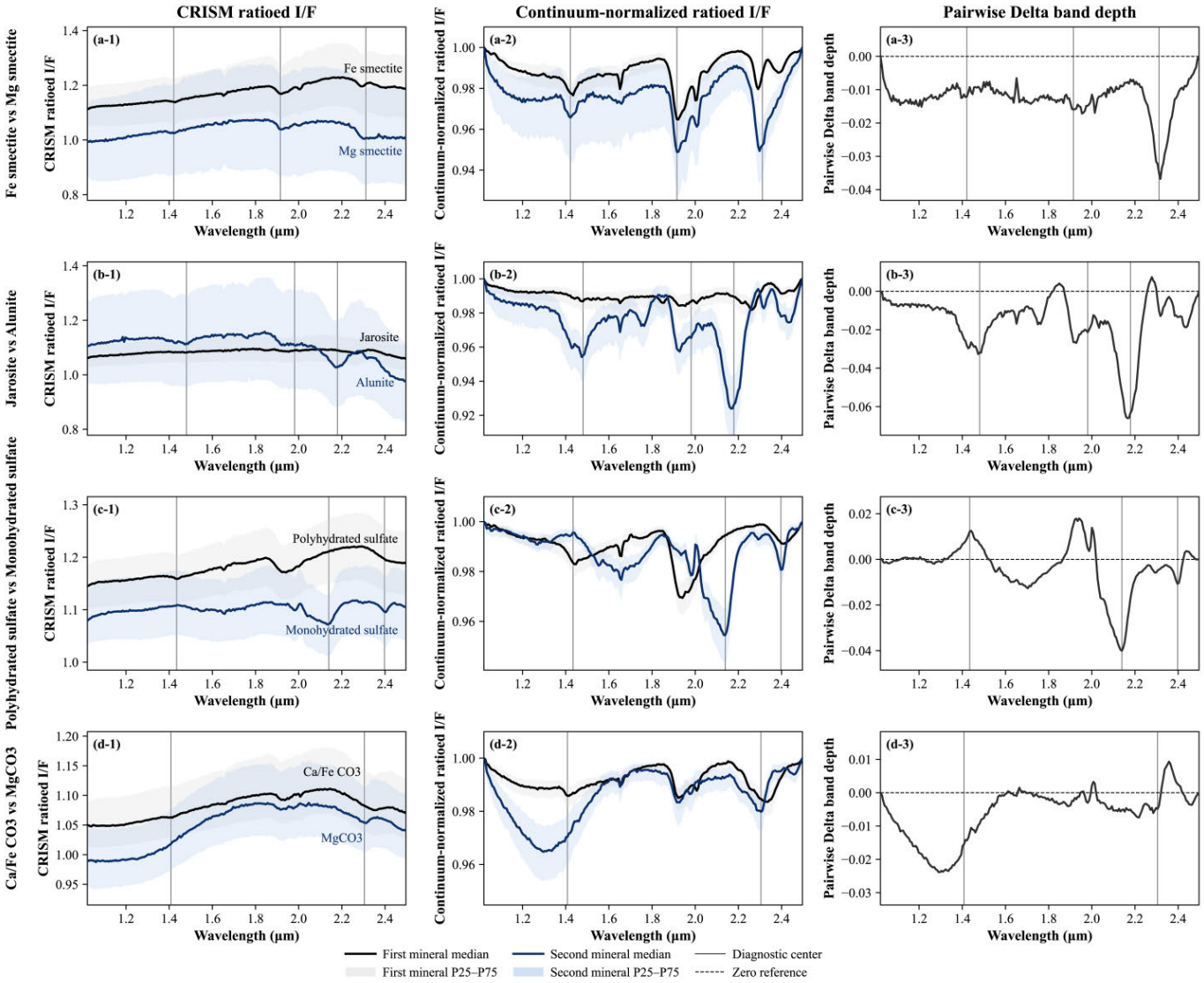


Fig. 18. Pairwise spectral discrimination for representative hard mineral pairs on the unified benchmark. (a-1)–(a-3) Fe-smectite versus Mg-smectite. (b1)–(b3) Jarosite versus alunite. (c1)–(c3) Polyhydrated sulfate versus monohydrated sulfate. (d1)–(d3) $CaCO_3$/$FeCO_3$ versus $MgCO_3$. For each mineral pair, the first, second, and third columns show CRISM ratioed I/F, continuum-normalized ratioed I/F, and pairwise Δ band depth, respectively.

fication performance, spatial mapping stability, and mineral-sensitive spectral discrimination.

Despite these advances, several limitations remain that define the scope of the current work. A major boundary of the current framework lies in the closed-set nature of the unified benchmark. While the unified benchmark provides a more standardized evaluation than scene-specific application tests, it remains inherently a closed-set supervised classification problem. The current results demonstrate VMSA's identification capability for a defined set of known target classes, but cannot be directly extrapolated to unknown-mineral discovery. The full CRISM archive contains far more complex unknown classes, transitional spectral morphologies, and under-annotated regions not represented in the closed benchmark, and the high accuracies reported here should be interpreted as performance on a standardized known-class benchmark, not as comprehensive coverage of realistic Martian mineral scenarios. Future work will incorporate label-uncertainty modeling, open-set recognition, and explicit rejection mechanisms to reduce overconfident predictions for unknown mineral classes and out-of-distribution spectral conditions [27], [28].

A second critical boundary is the handling of mixed pixels and complex mineral assemblages. While the spatial-consistency analysis with CRISM SSPs, the class-level spectral comparison in Fig. 7, and the pairwise continuum-normalized band-depth contrasts in Figs. 18 and 19 strengthen the correspondence between model decisions and mineral-sensitive absorption structure; they do not provide a complete solution to mixed-pixel interpretation. Many CRISM scenes are simultaneously affected by subpixel mixing, dust and basaltic background contamination, weathering coatings, and local topographic illumination variations, all of which can weaken, restructure, or even shift diagnostic absorption fea-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

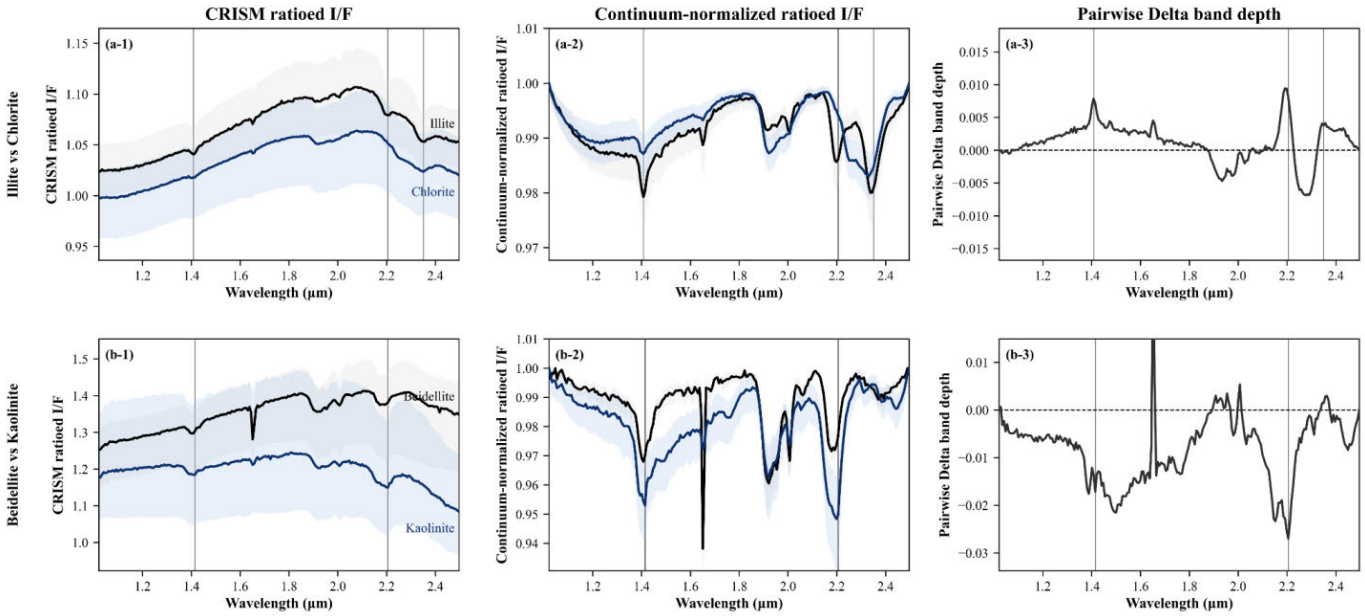


Fig. 19. Additional phyllosilicate hard-pair discrimination on the unified benchmark. (a1)–(a3) Illite versus chlorite. (b1)–(b3) Beidellite versus kaolinite. For each mineral pair, the first, second, and third columns show CRISM ratioed I/F, continuum-normalized ratioed I/F, and pairwise delta band depth, respectively.

tures. Accordingly, the present study is better interpreted as improving discrimination and mapping stability for spectrally similar mineral classes, rather than solving full compositional unmixing under mixed-pixel conditions. This boundary must remain explicit in all result interpretation and motivates caution when interpreting pixel-level classification results under mixed or strongly contaminated spectral conditions. Future work will integrate abundance-aware and subpixel-aware learning with physically constrained spectral-mixture modeling and regional geological validation, thereby extending closed-set pixel classification toward quantitative characterization of complex mineral assemblages [58], [59].

VI. Conclusion

In this article, we propose a VMSA framework for fine-grained classification and mapping of hydrated minerals from CRISM hyperspectral data. Aiming at the core limitations of existing pure data-driven methods, including over-reliance on local spectral–spatial features, poor discrimination of spectrally similar hydrated minerals, and limited mineralogical interpretability, VMSA adopts a dual-branch collaborative modeling architecture consisting of a CSB and a DPB. The framework integrates long-range spatial contextual learning with explicit prior modeling of mineral diagnostic absorption structures. The CSB learns local spectral–spatial structures and large-scale contextual continuity from stabilized hyperspectral cubes. The DPB explicitly embeds mineral-sensitive absorption-structure priors into the representation learning process through upper convex-hull continuum removal, group-specific diagnostic-interval extraction, sample-adaptive interval response, and class-conditional interval modulation, with joint discrimination of the two complementary feature streams achieved via the PGF module.

Based on the public CRISM machine learning resource, we construct a standardized unified benchmark for Martian hydrated-mineral recognition and systematically validate VMSA on three mineral-group subbenchmarks: phyllosilicates, sulfates, and carbonates. The results show that VMSA achieves the best overall performance among the compared methods on all three CRISM subbenchmarks under a unified training and testing protocol. Its advantage is not limited to a single mineral group or individual evaluation metric, but is reflected in overall classification accuracy, class-balanced recognition capability, and spatial consistency of classification results. These results demonstrate the stability and generalizability of the proposed method for fine-grained discrimination of spectrally similar hydrated-mineral classes.

Ablation experiments further reveal the core source of VMSA's performance gain. The improvement does not stem from incidental optimization of a single module, but from the structural synergy of the full workflow, including explicit diagnostic-prior modeling, prior-guided feature fusion, spectral-attention modeling, and long-range contextual modeling. The DPB plays a critical role in stable discrimination of spectrally similar minerals, with clear performance degradation observed across the three subbenchmarks when this branch is removed. The collaboration of the PGF module, SAM, and MCM further enables effective coupling between mineralogical priors and spectral–spatial contextual features, supporting the rationality of the overall architectural design.

Beyond benchmark-level quantitative evaluation, the results establish a consistent relationship between VMSA discrimination behavior and mineral-sensitive spectral structure. Cross-scene mapping comparisons demonstrate spatial coherence in representative CRISM observations, while CRISM SSPs provide regional mineralogical context for the predicted distributions. Fig. 7 further compares ratioed CRISM benchmark spectra with MICA-corrected and labo-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

ratory reference spectra for the 12 hydrated-mineral classes. Figs. 18 and 19 show that the principal hard mineral pairs retain stable continuum-normalized band-depth contrasts at selected diagnostic intervals. Together, these analyses indicate that the classification behavior of VMSA is associated with reproducible multiband spectral differences relevant to hydrated-mineral recognition.

The unified benchmark used in this work is inherently a closed-set supervised classification task, and its performance cannot be directly extrapolated to open-set scenarios for unknown-mineral discovery. In addition, although the class-level spectral context in Fig. 7 and the pairwise band-depth analyses in Figs. 18 and 19 strengthen the mineralogical interpretation of difficult hydrated-mineral classes, full compositional unmixing under mixed-pixel and complex mineral-assemblage conditions remains unresolved. Future work will focus on uncertainty-aware open-set recognition, abundance-aware subpixel spectral-mixture modeling, and large-scale validation across diverse geological settings in the CRISM archive.

ACKNOWLEDGMENT

The authors thank NASA for providing Compact Reconnaissance Imaging Spectrometer for Mars (CRISM) hyperspectral data.

REFERENCES

[1] J.-P. Bibring et al., "Global mineralogical and aqueous Mars history derived from OMEGA/Mars express data," *Science*, vol. 312, no. 5772, pp. 400–404, Apr. 2006, doi: 10.1126/science.1122659.

[2] S. Murchie et al., "Compact reconnaissance imaging spectrometer for Mars (CRISM) on Mars reconnaissance orbiter (MRO)," *J. Geophys. Res., Planets*, vol. 112, no. E5, 2007, Art. no. E05S03, doi: 10.1029/2006JE002682.

[3] B. L. Ehlmann et al., "Identification of hydrated silicate minerals on Mars using MRO-CRISM: Geologic context near Nili Fossae and implications for aqueous alteration," *J. Geophys. Res., Planets*, vol. 114, no. E2, Feb. 2009, Art. no. E00D08, doi: 10.1029/2009je003339.

[4] J. F. Mustard et al., "Hydrated silicate minerals on Mars observed by the Mars reconnaissance orbiter CRISM instrument," *Nature*, vol. 454, no. 7202, pp. 305–309, Jul. 2008, doi: 10.1038/nature07097.

[5] B. L. Ehlmann et al., "Subsurface water and clay mineral formation during the early history of Mars," *Nature*, vol. 479, no. 7371, pp. 53–60, Nov. 2011, doi: 10.1038/nature10582.

[6] J. Carter, L. Riu, F. Poulet, J.-P. Bibring, Y. Langevin, and B. Gondet, "A Mars orbital catalog of aqueous alteration signatures (MOCAAS)," *Icarus*, vol. 389, Jan. 2023, Art. no. 115164, doi: 10.1016/j.icarus.2022.115164.

[7] A. C. Allwood et al., "PIXL: Planetary instrument for X-ray lithochemistry," *Space Sci. Rev.*, vol. 216, no. 8, p. 134, Dec. 2020, doi: 10.1007/s11214-020-00767-7.

[8] K. L. Siebach, E. L. Moreland, G. Costin, and Y. Jiang, "MIST: An online tool automating mineral identification by stoichiometry," *Comput. Geosci.*, vol. 206, Jan. 2026, Art. no. 106021, doi: 10.1016/j.cageo.2025.106021.

[9] E. L. Moreland et al., "Multiple episodes of fluid alteration in Jezero Crater indicated by MIST mineral identifications in PIXL XRF data from the first 1100 sols of the Mars 2020 mission," *J. Geophys. Res., Planets*, vol. 130, no. 9, p. 2024, Sep. 2025, doi: 10.1029/2024je008797.

[10] Z. He et al., "Mars mineralogical spectrometer (MMS) on the Tianwen-1 mission," *Space Sci. Rev.*, vol. 217, no. 2, p. 27, Mar. 2021, doi: 10.1007/s11214-021-00804-z.

[11] P. C. McGuire et al., "An improvement to the volcano-scan algorithm for atmospheric correction of CRISM and OMEGA spectral data," *Planet. Space Sci.*, vol. 57, no. 7, pp. 809–815, Jun. 2009, doi: 10.1016/j.pss.2009.03.007.

[12] S. M. Pelkey et al., "CRISM multispectral summary products: Parameterizing mineral diversity on Mars from reflectance," *J. Geophys. Res., Planets*, vol. 112, no. E8, 2007, Art. no. E08S14, doi: 10.1029/2006JE002831.

[13] C. E. Viviano et al., "Revised CRISM spectral parameters and summary products based on the currently detected mineral diversity on Mars," *J. Geophys. Res., Planets*, vol. 119, no. 6, pp. 1403–1431, Jun. 2014, doi: 10.1002/2014JE004627.

[14] Y. Itoh and M. Parente, "A new method for atmospheric correction and de-noising of CRISM hyperspectral data," *Icarus*, vol. 354, Jan. 2021, Art. no. 114024, doi: 10.1016/j.icarus.2020.114024.

[15] B. Hapke, "Bidirectional reflectance spectroscopy: 1. Theory," *J. Geophys. Res., Solid Earth*, vol. 86, no. B4, pp. 3039–3054, Apr. 1981.

[16] R. N. Clark, "Spectroscopy of rocks and minerals, and principles of spectroscopy," in *Infrared Spectro Scopy in Geochemistry, Exploration Geochemistry and Remote Sensing* (Short Course Series), vol. 33, P. L. King, M. S. Ramsey, and G. A. Swayze, Eds., Ottawa, ON, Canada: Mineralogical Association of Canada, 2004, pp. 17–55, doi: 10.3749/9780921294771.ch02.

[17] E. Plebani, B. L. Ehlmann, E. K. Leask, V. K. Fox, and M. M. Dundar, "A machine learning toolkit for CRISM image analysis," *Icarus*, vol. 376, Apr. 2022, Art. no. 114849, doi: 10.1016/j.icarus.2021.114849.

[18] A. M. Saranathan and M. Parente, "Adversarial feature learning for improved mineral mapping of CRISM data," *Icarus*, vol. 355, Feb. 2021, Art. no. 114107, doi: 10.1016/j.icarus.2020.114107.

[19] P. Kumari, S. Soor, A. Shetty, and S. G. Koolagudi, "A fully-automated framework for mineral identification on Martian surface using supervised learning models," *IEEE Access*, vol. 11, pp. 13121–13137, 2023, doi: 10.1109/ACCESS.2023.3243061.

[20] F. P. Seelos et al., "The CRISM investigation in Mars orbit: Overview, history, and delivered data products," *Icarus*, vol. 419, Sep. 2024, Art. no. 115612, doi: 10.1016/j.icarus.2023.115612.

[21] R. Clark et al., "USGS digital spectral library splib06a," *Data Series*, Reston, VA, USA, Tech. Rep. 231, 2007. [Online]. Available: https://pubs.usgs.gov/publication/ds231

[22] R. E. Milliken, T. Hiroi, D. Scholes, S. Slavney, and R. Arvidson, "The NASA reflectance experiment LABoratory (RELAB) facility: An online spectral database for planetary exploration," in *Proc. Astromaterials Data Manag. Era Sample-Return Missions Community Workshop*, 2021. [Online]. Available: https://ui.adsabs.harvard.edu/abs/2021LPICo2654.2021M

[23] T. Ke, Y. Zhong, M. Song, X. Wang, and L. Zhang, "Mineral detection based on hyperspectral remote sensing imagery on Mars: From detection methods to fine mapping," *ISPRS J. Photogramm. Remote Sens.*, vol. 218, pp. 761–780, Dec. 2024, doi: 10.1016/j.isprsjprs.2024.09.020.

[24] P. Kumari, S. Soor, A. Shetty, and S. G. Koolagudi, "MICAnet: A deep convolutional neural network for mineral identification on Martian surface," *Egyptian J. Remote Sens. Space Sci.*, vol. 27, no. 3, pp. 501–507, Sep. 2024, doi: 10.1016/j.ejrs.2024.06.001.

[25] P. Kumari, S. Soor, A. Shetty, and A. M. Nair, "A UNet model for accelerated preprocessing of CRISM hyperspectral data for mineral identification on Mars," 2025, *arXiv:2505.02046*.

[26] D. Hürland, A. Pletl, M. Fernandes, and B. Elser, "Framework for processing of CRISM hyperspectral data for global Martian mineralogy," *Remote Sens.*, vol. 17, no. 23, p. 3831, Nov. 2025, doi: 10.3390/rs17233831.

[27] S. Dhoundiyal, M. S. Dey, S. Singh, P. V. Arun, G. Thangjam, and A. Porwal, "Open-set identification of minerals from CRISM hyperspectral data," in *Proc. IGARSS - IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2024, pp. 6164–6167, doi: 10.1109/IGARSS53475.2024.10642309.

[28] S. Dhoundiyal, M. S. Dey, S. Singh, P. V. Arun, G. Thangjam, and A. Porwal, "Explainable machine learning for mapping minerals from CRISM hyperspectral data," *J. Geophys. Res., Mach. Learn. Comput.*, vol. 2, no. 2, p. 2024, Jun. 2025, doi: 10.1029/2024jh000391.

[29] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014, doi: 10.1109/JSTARS.2014.2329330.

[30] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615, doi: 10.1109/TGRS.2021.3130716.

[31] K. Wang, X. Zhang, X. Wang, and Z. Qian, "Identification of Martian minerals based on multiscale spatial–spectral fusion network," *Eng. Appl. Artif. Intell.*, vol. 149, Jun. 2025, Art. no. 110527, doi: 10.1016/j.engappai.2025.110527.

[32] B. Xi et al., "MCTGCL: Mixed CNN–transformer for Mars hyperspectral image classification with graph contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5503214, doi: 10.1109/TGRS.2025.3529996.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.

[33] S. Liu et al., "SHMD: A sequential approach for Martian hydrated minerals detection and abundance mapping," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 1023–1037, 2026, doi: 10.1109/JSTARS.2025.3637770.

[34] P. R. Christensen et al., "Mars global surveyor thermal emission spectrometer experiment: Investigation description and surface science results," *J. Geophys. Res., Planets*, vol. 106, no. E10, pp. 23823–23871, Oct. 2001.

[35] J.-P. Bibring et al., "Mars surface diversity as revealed by the OMEGA/Mars express observations," *Science*, vol. 307, no. 5715, pp. 1576–1581, Mar. 2005, doi: 10.1126/science.1108806.

[36] Y. Zou et al., "Scientific objectives and payloads of Tianwen-1, China's first Mars exploration mission," *Adv. Space Res.*, vol. 67, no. 2, pp. 812–823, Jan. 2021, doi: 10.1016/j.asr.2020.11.005.

[37] S. L. Murchie et al., "Compact reconnaissance imaging spectrometer for Mars investigation and data set from the Mars reconnaissance Orbiter's primary science phase," *J. Geophys. Res., Planets*, vol. 114, no. E2, Feb. 2009, Art. no. E00D07, doi: 10.1029/2009je003344.

[38] B. Liu et al., "Ground validation experiment and spectral detection capability evaluation of Mars mineralogical spectrometer (MMS) aboard HX-1 orbiter," *Space Sci. Rev.*, vol. 218, Feb. 2022, Art. no. 1, doi: 10.1007/s11214-021-00858-z.

[39] B. Wu, R. Xu, C. Liu, and Z. He, "In-orbit detection of the spectral smile for the Mars mineral spectrometer," *ISPRS J. Photogramm. Remote Sens.*, vol. 216, pp. 32–44, Oct. 2024, doi: 10.1016/j.isprsjprs.2024.07.023.

[40] B. L. Ehlmann, J. F. Mustard, R. N. Clark, G. A. Swayze, and S. L. Murchie, "Evidence for low-grade metamorphism, hydrothermal alteration, and diagenesis on Mars from phyllosilicate mineral assemblages," *Clays Clay Minerals*, vol. 59, no. 4, pp. 359–377, Aug. 2011, doi: 10.1346/ccmn.2011.0590402.

[41] J. L. Bishop and E. B. Rampe, "Evidence for a changing Martian climate from the mineralogy at Mawrth Vallis," *Earth Planet. Sci. Lett.*, vol. 448, pp. 42–48, Aug. 2016, doi: 10.1016/j.epsl.2016.04.031.

[42] A. Gendrin et al., "Sulfates in Martian layered terrains: The OMEGA/Mars express view," *Science*, vol. 307, no. 5715, pp. 1587–1591, Mar. 2005, doi: 10.1126/science.1109087.

[43] I. C. Das, J. Joseph, S. K. Subramanian, and V. K. Dadhwal, "Spectral stratigraphy and clay minerals analysis in parts of Hellas Planitia, Mars," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 40, pp. 419–422, Nov. 2014, doi: 10.5194/isprsarchives-xl-8-419-2014.

[44] K. A. Lichtenberg et al., "Stratigraphy of hydrated sulfates in the sedimentary deposits of Aram chaos, Mars," *J. Geophys. Res., Planets*, vol. 115, no. E6, Jun. 2010, Art. no. E00D17, doi: 10.1029/2009je003353.

[45] J. J. Wray et al., "Orbital evidence for more widespread carbonate-bearing rocks on Mars," *J. Geophys. Res., Planets*, vol. 121, no. 4, pp. 652–677, Apr. 2016, doi: 10.1002/2015je004972.

[46] R. E. Milliken et al., "Opaline silica in young deposits on Mars," *Geology*, vol. 36, no. 11, pp. 847–850, 2008, doi: 10.1130/g24967a.1.

[47] M. Pineau, L. Le Deit, B. Chauviré, J. Carter, B. Rondeau, and N. Mangold, "Toward the geological significance of hydrated silica detected by near infrared spectroscopy on Mars based on terrestrial reference samples," *Icarus*, vol. 347, Sep. 2020, Art. no. 113706, doi: 10.1016/j.icarus.2020.113706.

[48] E. A. Cloutis, P. M. Asher, and S. A. Mertzman, "Spectral reflectance properties of zeolites and remote sensing implications," *J. Geophys. Res., Planets*, vol. 107, no. E9, pp. 1–19, Sep. 2002, doi: 10.1029/2000je001467.

[49] C. E. Viviano-Beck et al., "Revised CRISM spectral parameters and summary products," in *Proc. 45th Lunar Planet. Sci. Conf.*, The Woodlands, TX, USA, Mar. 2014.

[50] A. A. Green, M. Berman, P. Switzer, and M. D. Craig, "A transformation for ordering multispectral data in terms of image quality with implications for noise removal," *IEEE Trans. Geosci. Remote Sens.*, vol. 26, no. 1, pp. 65–74, Jan. 1988, doi: 10.1109/36.3001.

[51] R. N. Clark and T. L. Roush, "Reflectance spectroscopy: Quantitative analysis techniques for remote sensing applications," *J. Geophys. Res., Solid Earth*, vol. 89, no. B7, pp. 6329–6340, Jul. 1984, doi: 10.1029/jb089ib07p06329.

[52] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, and S. Belongie, "Class-balanced loss based on effective number of samples," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9260–9269.

[53] W. Liu, Y. Wen, Z. Yu, and M. Yang, "Large-margin softmax loss for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2016, pp. 1–11.

[54] M. Morgan, F. Seelos, and S. Murchie, "The CRISM analysis toolkit (CAT): Overview and recent updates," in *Proc. 3rd Planet. Data Workshop Planet. Geologic Mappers Annu. Meeting*, 2017, p. 7121.

[55] S. K. Roy, S. Manna, T. Song, and L. Bruzzone, "Attention-based adaptive spectral–spatial kernel ResNet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7831–7843, Sep. 2021.

[56] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214, doi: 10.1109/TGRS.2022.3144158.

[57] Y. Li, Y. Luo, L. Zhang, Z. Wang, and B. Du, "MambaHSI: Spatial–spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5524216, doi: 10.1109/TGRS.2024.3430985.

[58] Y. Liu et al., "End-member identification and spectral mixture analysis of CRISM hyperspectral data: A case study on Southwest Melas Chasma, Mars," *J. Geophys. Res., Planets*, vol. 121, no. 10, pp. 2004–2036, Oct. 2016, doi: 10.1002/2016je005028.

[59] H. Lin and X. Zhang, "Retrieving the hydrous minerals on Mars by sparse unmixing and the Hapke model using MRO/CRISM data," *Icarus*, vol. 288, pp. 160–171, May 2017, doi: 10.1016/j.icarus.2017.01.019.



Yanchao Zheng (Graduate Student Member, IEEE) received the M.S. degree in surveying and mapping science and technology from Henan Polytechnic University, Jiaozuo, China, in 2023, and currently he is currently pursuing the Ph.D. degree with the School of Geomatics and Aerospace Information, Henan Polytechnic University, Jiaozuo, China.

His research interests include planetary mapping and remote sensing.



Shuanggen Jin (Senior Member, IEEE) was born in Anhui, China, in 1974. He received the B.Sc. degree in geodesy from Wuhan University, Wuhan, China, in 1999, and the Ph.D. degree in geodesy from the University of Chinese Academy of Sciences, Beijing, China, in 2003.

He is currently the Vice-President and a Professor with Henan Polytechnic University, Jiaozuo, China, and also a Professor with the School of Artificial Intelligence, Anhui University, Hefei, China. He has published over 500 articles in peer-reviewed journals and proceedings, 50 patents/software copyrights, and 20 books/monographs with more than 20 000 citations and an H-index > 77. His main research areas include satellite navigation, remote sensing, and space/planetary exploration.

Prof. Jin was a fellow of IAG, IUGG, and the Electromagnetics Academy; a World Class Professor of the Ministry of Education and Culture, Indonesia; the Chief Scientist of the National Key Research and Development Program of China; and a member of Russian Academy of Natural Sciences, European Academy of Sciences, Turkish Academy of Sciences, and Academia Europaea. He was the President of the International Association of Planetary Sciences (IAPS) from 2015 to 2019 and the International Association of CPGNSS from 2016 to 2017. He received the 100-Talent Program of CAS and Leading Talent of Shanghai. He is the Chair of the IUGG Union Commission on Planetary Sciences (UCPS) for the period 2015–2027. He was the Editor-in-Chief of the *International Journal of Geosciences*, an Editor of the *Geoscience Letters*, an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and *Journal of Navigation*, and an Editorial Board Member of *Remote Sensing*, *GPS Solutions*, and *Journal of Geodynamics*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:51:09 UTC from IEEE Xplore. Restrictions apply.